

Longitudinal assessment of DREADD expression and efficacy in the monkey brain

Reviewed Preprint

v2 • September 24, 2025

Revised by authors

Reviewed Preprint

v1 • April 2, 2025

Yuji Nagai , Yukiko Hori, Ken-ichi Inoue, Toshiyuki Hirabayashi, Koki Mimura, Kei Oyama, Naohisa Miyakawa, Yuki Hori, Haruhiko Iwaoki, Katsushi Kumata, Ming-Rong Zhang, Masahiko Takada, Makoto Higuchi, Takafumi Minamimoto 

Advanced Neuroimaging Center, National Institutes for Quantum Science and Technology, Chiba, Japan • System Neuroscience Section, Center for the Evolutionary Origins of Human Behavior, Kyoto University, Inuyama, Japan • Department of Advanced Nuclear Medicine Sciences, National Institutes for Quantum Science and Technology, Chiba, Japan

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This study provides novel and **fundamental** insights into the long-term use of DREADDs to modulate neuronal activity in nonhuman primates. The **exceptional** evidence demonstrates the peak dynamics and the subsequent stability of chemogenetic effects for 1.5 years, informing the experimental designs and the interpretation of highly impactful chemogenetic studies in macaques. The protocols, data, and outcomes can serve as guidelines for future experiments. Therefore, the findings will be of significant interest to the field of chemogenetics and may also be of broader interest to researchers and clinicians who seek to utilize viral vectors and/or related genetic technologies.

<https://doi.org/10.7554/eLife.105815.2.sa4>

Abstract

Designer Receptors Exclusively Activated by Designer Drugs (DREADDs) offer a powerful means for reversible control of neuronal activity through systemic administration of inert actuators. Because chemogenetic control relies on DREADD expression levels, understanding and quantifying the temporal dynamics of their expression is crucial for planning long-term experiments in monkeys. In this study, we longitudinally quantified *in vivo* DREADD expression in macaque monkeys using positron emission tomography with the DREADD-selective tracer [¹¹C]deschloroclozapine (DCZ), complemented by functional studies. Twenty macaque monkeys were evaluated after being injected with adeno-associated virus vectors expressing the DREADDs hM4Di or hM3Dq, whose expression was quantified as changes in [¹¹C]DCZ binding potential from baseline levels. Expression levels of both hM4Di and hM3Dq peaked around 60 days post-injection, remained stable for about 1.5 years, and declined gradually after two years. Significant chemogenetic control of neural activity and behavior persisted for about two years. The presence of protein tags significantly influenced expression levels, with co-expressed protein tags reducing overall expression levels. These

findings provide valuable insights and guidelines for optimizing the use of DREADDs in long-term primate studies and potential therapeutic applications.

Introduction

Chemogenetic technology affords remote and reversible control of neuronal activity by expressing receptors that are designed to be activated by systemically delivered biologically inert actuators. One such tool is the widely used Designer Receptors Exclusively Activated by Designer Drugs (DREADD) system, which is based on mutated muscarinic receptors (Roth, 2016 [↗](#)). DREADDs have been widely used in rodent models and their use in nonhuman primates has recently shown some promise by inducing significant behavioral changes through simultaneous modulation of activity in disparate neuronal populations across large brain regions (Eldridge et al., 2016; Nagai et al., 2016 [↗](#)). Whether this promise will translate to reality depends on how long DREADDs can exert their effects. Because the efficacy of chemogenetic manipulation depends on DREADD expression levels (Grayson et al., 2016 [↗](#); Upright et al., 2018 [↗](#)), maintaining high and stable expression levels is critical for ensuring consistent control throughout all experiments in a study. While this has proven possible in rodent studies that typically only span several months, whether it is true for studies involving macaque monkeys, which often last multiple years, remains a question. Indeed, despite the growing use of DREADDs in monkey neuroscience research, consistent and reliable data on the temporal dynamics of DREADD expression following viral transduction remain scarce. Specifically, little is known about how quickly DREADD expression is established following viral transduction or how long it is maintained. This gap in knowledge complicates efforts to optimize critical parameters, such as the serotypes, constructs, and titers of viral vectors that all influence expression level and duration.

Positron emission tomography (PET) is an *in vivo* imaging technique that allows quantification of receptor density in the brain. Several studies have demonstrated the utility of PET for longitudinal measurements of gene expression introduced into the brain via adeno-associated virus (AAV) (Nerella et al., 2023 [↗](#); Vandeputte et al., 2011 [↗](#); Yoon et al., 2014 [↗](#)). While these studies have highlighted the potential of using PET imaging to monitor gene expression over time, most have used it as an indirect measure of target gene expression, without exploring the relationship between expression levels and the resultant functional outcomes.

Several PET probes have been developed to visualize DREADD receptor expression. Initial studies in rodents used [^{11}C]clozapine-N-oxide (CNO) and [^{11}C]clozapine to monitor DREADD expression in the brain (Ji et al., 2016 [↗](#)). In nonhuman primates, longitudinal PET studies using [^{11}C]clozapine have successfully tracked virally expressed DREADDs in macaques (Nagai et al., 2016 [↗](#)). More recently, DREADD-selective ligands such as [^{18}F]JHU37107 (Bonaventura et al., 2019 [↗](#)) and [^{11}C]deschloroclozapine ([^{11}C]DCZ) (Nagai et al., 2020 [↗](#)) have been developed. Among these, [^{11}C]DCZ has become a particularly useful tracer for studying DREADD-mediated functional manipulation in macaques and marmosets. This PET ligand allows researchers to verify the location and density of DREADD expression at the site of AAV injection and at projection sites, as demonstrated in numerous studies (Hirabayashi et al., 2024 [↗](#), 2021 [↗](#); Hori et al., 2021 [↗](#); Mimura et al., 2021 [↗](#); Miyakawa et al., 2023 [↗](#); Miyake et al., 2024 [↗](#); Mueller et al., 2023 [↗](#); Nagai et al., 2020 [↗](#), 2016 [↗](#); Oyama et al., 2024 [↗](#), 2022a [↗](#), 2022b [↗](#), 2021 [↗](#); Roseboom et al., 2021 [↗](#)). Moreover, unlike post-mortem analysis that relies on the detection of fused or expressed protein tags, [^{11}C]DCZ PET provides a direct, quantitative, and non-invasive measure of DREADD expression level *in vivo*, using standard reference tissue models (Nagai et al., 2020 [↗](#); Yan et al., 2021 [↗](#)). [^{11}C]DCZ PET thus serves as a valuable tool for tracking DREADD expression levels throughout long-term chemogenetics experiments in monkeys.

As the developer of the DREADD-selective agonist DCZ (Nagai et al., 2020 [DOI](#)), we have consistently performed multiple [^{11}C]DCZ PET scans as part of our DREADD experiments in monkeys, including at baseline before AAV injection, 30–120 days post-injection, and then periodically over the course of the experiments. For the current study, we analyzed these PET data to examine the short- and long-term dynamics of DREADD expression *in vivo* and assessed how these changes correlated with the ability to manipulate neuronal activity or behavior. Our results provide valuable information with a few caveats and indicate that the promise of using DREADDs in long-term monkey experiments is real.

Results

PET imaging data using [^{11}C]DCZ were collected from 15 monkeys before (baseline) and after injections of AAV vectors for expressing hM4Di or hM3Dq (see **Table 1** [DOI](#)). To achieve neuron-specific expression, we used AAV vectors with a preference for neuronal infection, such as AAV2 or AAV2.1 (Kimura et al., 2023 [DOI](#)), or for incorporation with neuron-specific promoters (e.g., human synapsin promoter; hSyn). To assess the specific binding of [^{11}C]DCZ, we calculated the binding potential relative to a non-displaceable radioligand (BP_{ND}) using a multilinear reference tissue model with the cerebellum as a reference region (Yan et al., 2021 [DOI](#)). DREADD expression levels were then quantified as the change in BP_{ND} ($\Delta\text{BP}_{\text{ND}}$) from baseline values. This approach allowed us to longitudinally and quantitatively monitor the dynamics of DREADD expression *in vivo*.

Peak DREADD expression occurred approximately 60 days post-injection

To evaluate the temporal profile of DREADD expression and determine how quickly expression was established following transduction, we analyzed the [^{11}C]DCZ PET data from seven monkeys that received AAV vector injections into subcortical regions with similar volumes (**Tables 1** [DOI](#) and **2** [DOI](#)). These data were originally collected for vector-testing studies, during which PET scans were frequently repeated up to 150 days post-injection. The dataset was selected according to the following criteria: (1) the titer of injected viral vector was $1.0\text{--}3.0 \times 10^{13}$ gc/ml, and (2) the peak $\Delta\text{BP}_{\text{ND}}$ value exceeded 0.5. Analysis showed that expression levels of hM4Di and hM3Dq increased rapidly, reaching peak expression at approximately 60 days after viral vector injection. Notably, no clear differences in expression dynamics were observed between hM3Dq and hM4Di (**Figure 1** [DOI](#)).

Longitudinal assessment of DREADD expression levels

We then conducted a longitudinal assessment of post-peak expression levels for hM4Di and hM3Dq using repeated [^{11}C]DCZ PET measurements from 16 injection sites, including cortical and subcortical regions, across 11 monkeys. **Figure 2** [DOI](#) illustrates the temporal changes in normalized $\Delta\text{BP}_{\text{ND}}$, expressed as a percentage of the peak value observed 40–80 days post-AAV injection. hM4Di expression levels remained stable at peak levels for approximately 1.5 years, followed by a gradual decline observed in one case after 2.5 years, and after approximately 3 years in the other two cases (**Figure 2B** [DOI](#), o and b/e, respectively). Compared with hM4Di expression, hM3Dq expression exhibited greater post-peak fluctuations. Nevertheless, it remained at ~70% of peak levels after about 1 year. This post-peak fluctuation was not significantly associated with the cumulative number of DREADD agonist injections (repeated-measures two-way ANOVA, main effect of activation times, $F_{(1,6)} = 5.745$, $P = 0.054$). Beyond 2 years post-injection, expression declined to ~50% in one case, whereas another case showed an apparent increase (**Figure 2C** [DOI](#), c and m, respectively).

ID	Species	Sex	weight (kg)	age (y)	DREADD	[¹¹ C]DCZ	Behavior	FDG	Electrophysiology
153	R	M	6.8	10	hM4Di		✓ ^{*1)}		
163	R	M	6.1	13	hM4Di		✓ ^{*2)}		
193	R	M	4.4	9	hM4Di	✓			
201	R	M	6.3	8	hM4Di		✓ ^{*3)}		✓
207	R	M	6.8	6	hM4Di		✓ ^{*3)}		✓
212	R	F	3.4	8	hM4Di	✓			
218	J	M	7.2	4	hM4Di		✓ ^{*4)}		
221	J	M	6.4	4	hM4Di	✓	✓ ^{*1)}		
225	J	F	6	7	hM4Di	✓	✓ ^{*5)}		
229	R	M	4.6	4	hM4Di	✓	✓ ^{*6)}		
234	J	M	5.8	5	hM4Di	✓	✓ ^{*5)}		
237	J	M	6.9	5	hM4Di	✓			
238	J	M	6.2	4	hM4Di	✓	✓ ^{*4)}		
245	J	F	5.9	6	hM4Di	✓	✓ ^{*6)}		
215	R	F	3.4	8	hM3Dq	✓		✓	
223	C	M	4.2	4	hM3Dq	✓		✓	
224	J	M	9.9	9	hM3Dq	✓			✓
236	J	M	7.1	6	hM3Dq	✓		✓	
241	J	M	5.1	3	hM3Dq	✓		✓	
255	C	M	5.2	4	hM3Dq	✓		✓	

R, Rhesus; J, Japanese; C, Cynomolgus; M, male; F, female. Weight and age are the values recorded at the beginning of the experiments. Behavior: *1) multi-reward task, #153 and #221, Oyama et al., *J Neurosci* 2022; *2) delayed-reward task, #163 Hori et al., *eLife* 2021; *3) delayed matching-to-sample task, #201 and #207, Hirabayashi et al., *Nat Comm* 2024; *4) reversal learning task, #218 and #238, Oyama et al., *Nat Comm* 2024; *5) Brinkman board/Foot sensation, #225 and #234, Hirabayashi et al., *Neuron* 2021; *6) delayed response task, #229 and #245, Nagai et al., *Nat Neurosci* 2020.

Table 1.

Summary of subjects, DREADD used, and functional assessments used in this study

ID	Region	Vector	Titer ($\times 10^{13}$ gc/ml)	Volume (μ l)	Symbols in figures			
					1	2	5	S1
153	R-OFC	AAV2-CMV-hM4Di	1.0	54				
	L-rmCD	AAV2-CMV-hM4Di	2.0	3				
163	L-dCDh	AAV2.1-hSyn-hM4Di-IRES2-AcGFP	4.7	6				
	R-dCDh	AAV2.1-hSyn-hM4Di-IRES2-AcGFP	4.7	6				
193	R-Amygdala	AAV2-CMV-hM4Di	2.0	6	a		a	a
201	R-OFC (microinjected)	AAV2-CMV-hM4Di	2.2	2				
	L-OFC (microinjected)	AAV2-CMV-hM4Di	2.2	2				
207	R-OFC (microinjected)	AAV2-CMV-hM4Di	1.3	3				
	L-OFC (microinjected)	AAV2-CMV-hM4Di	1.3	3				
212	R-Put	AAV2-CMV-hM4Di	2.6	6	b	b	b	b
215	L-Amygdala	AAV2-CMV-hM3Dq	1.2	6	c	c	c	c
218	R-OFC	AAV2-CMV-hM4Di	2.3	50				
	L-OFC	AAV2-CMV-hM4Di	2.3	54				
221	L-OFC	AAV2-CMV-hM4Di	2.0	50				
	R-rmCD	AAV2-CMV-hM4Di	2.0	3		k	k	k
223	R-Cd	AAV2.1-hSyn-hM3Dq-IRES2-AcGFP	1.0	3	d	d	d	d
	R-Put	AAV2.1-hSyn-hM3Dq-IRES2-AcGFP	5.0	3		l		
	L-Cd	AAV2-CMV-hM3Dq-IRES-AcGFP	1.0	3				
	L-Put	AAV2-hSyn-hM3Dq-IRES2-AcGFP	1.0	3			r	r
224	R-Amygdala	AAV2.1-hSyn-hM3Dq-IRES-AcGFP	2.0	5		m	m	m
	L-Amygdala	AAV2.1-hSyn-hM3Dq-IRES-AcGFP	2.0	4			s	s
225	L-SID ₂	AAV2-CMV-hM4Di	1.5	4			t	t
229	R-dlPFC	AAV2.1-hSyn-hM4D-IRES2-AcGFP	4.7	35		n		
	L-dlPFC	AAV2.1-hSyn-hM4D-IRES2-AcGFP	4.7	37				
234	R-SID ₂	AAV2.1-hSyn-hM4D-IRES2-AcGFP	3.8	4		o	o	o
236	R-Amygdala	AAV2-CMV-hM3Dq	1.2	6				
237	L-Amygdala	AAV2.1-hSyn-hM4Di-IRES2-AcGFP	2.0	4	e	e	e	e

Table 2.

Summary of injection location, type of virus vector, titer, and injected volume used in this study

238	R-OFC	AAV2.1-CaMKII-hM4Di-IRES-AcGFP	1.0	49	q			
	L-OFC	AAV2.1-CaMKII-hM4Di-IRES-AcGFP	1.0	53	p			
241	R-Cd	AAV2-hSyn-hM3Dq-IRES-AcGFP	2.0	3	u u			
	L-Cd	AAV2-hSyn-hM3Dq	2.0	3	g	g	g	g
	R-Put	AAV2.1-hSyn-hM3Dq-IRES2-AcGFP	2.0	3	h	h	h	h
	L-Put	AAV1-hSyn-hM3Dq-IRES2-AcGFP	2.0	3	f	f	f	f
245	R-dlPFC	AAV2.1-hSyn-hM4D-IRES2-AcGFP	4.7	44				
	L-dlPFC	AAV2.1-hSyn-hM4D-IRES2-AcGFP	4.7	40				
255	L-VPL	AAV5-hSyn-HA-hM3Dq	2.7	3	j	j	j	j
	R-VPL	AAV2-hSyn-hM3Dq	1.8	3	i	i	i	i
	L-Cd	AAV2-hSyn-hM4Di	1.2	3	x x			
	R-Cd	AAV2-hSyn-hM4Di	1.7	3	v v			
	L-Put	AAV5-hSyn-hM4Di	4.6	3				
	R-Put	AAV2-hSyn-hM4Di	1.7	3	w w			

R, right; *L*, left; *OFC*, orbitofrontal cortex; *rmCD*, rostromedial caudate nucleus; *dCDh*, dorsal part of caudate nucleus head; *CD*, caudate nucleus; *Put*, putamen; *SI_{D2}*, hand index finger region of primary somatosensory cortex; *dlPFC*, dorsolateral prefrontal cortex; *VPL*, ventral posterolateral nucleus of thalamus.

Table 2. (continued)

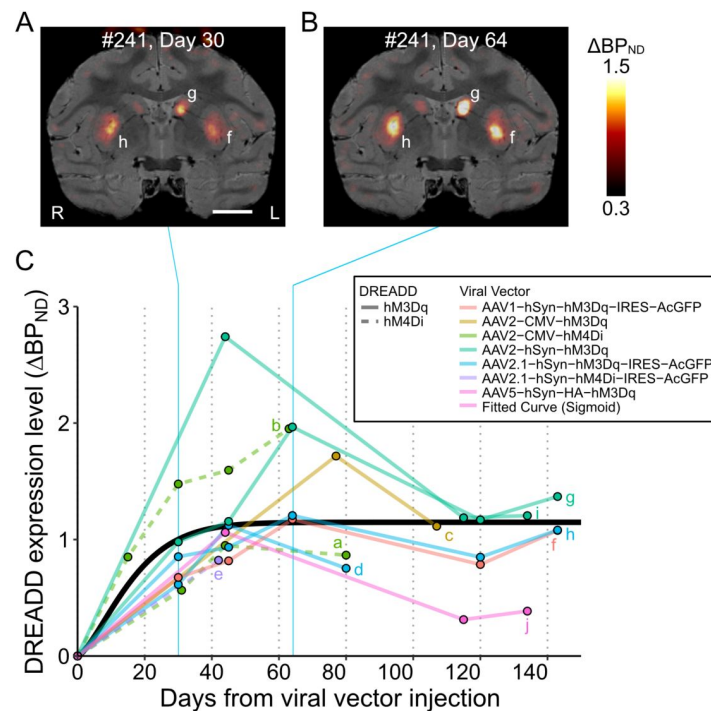


Figure 1

Time course of DREADD expression levels up to 150 days after injection

(A, B) Parametric coronal images of the increase (relative to baseline) in specific binding of $[^{11}\text{C}]\text{DCZ}$ ($\Delta\text{BP}_{\text{ND}}$) at 30 (A) and 64 (B) days after AAV injection, overlaid on MR images. Images were obtained from a monkey (#241) that received multiple injections of AAV with different constructs (see Table 2). (C) Time course of *in vivo* DREADD expression levels ($\Delta\text{BP}_{\text{ND}}$) up to 150 days after the injections, summarized from 10 regions of interest (ROIs) obtained from 7 monkeys. The value at day 0 indicates the baseline (before injection). The black curve is the best-fitted sigmoid curve, which provides a better fit (Bayesian Information Criterion, BIC = 61.1) than does the double logistic model (BIC = 62.9). Lowercase letters correspond to the DREADD-induced regions described in Table 2.

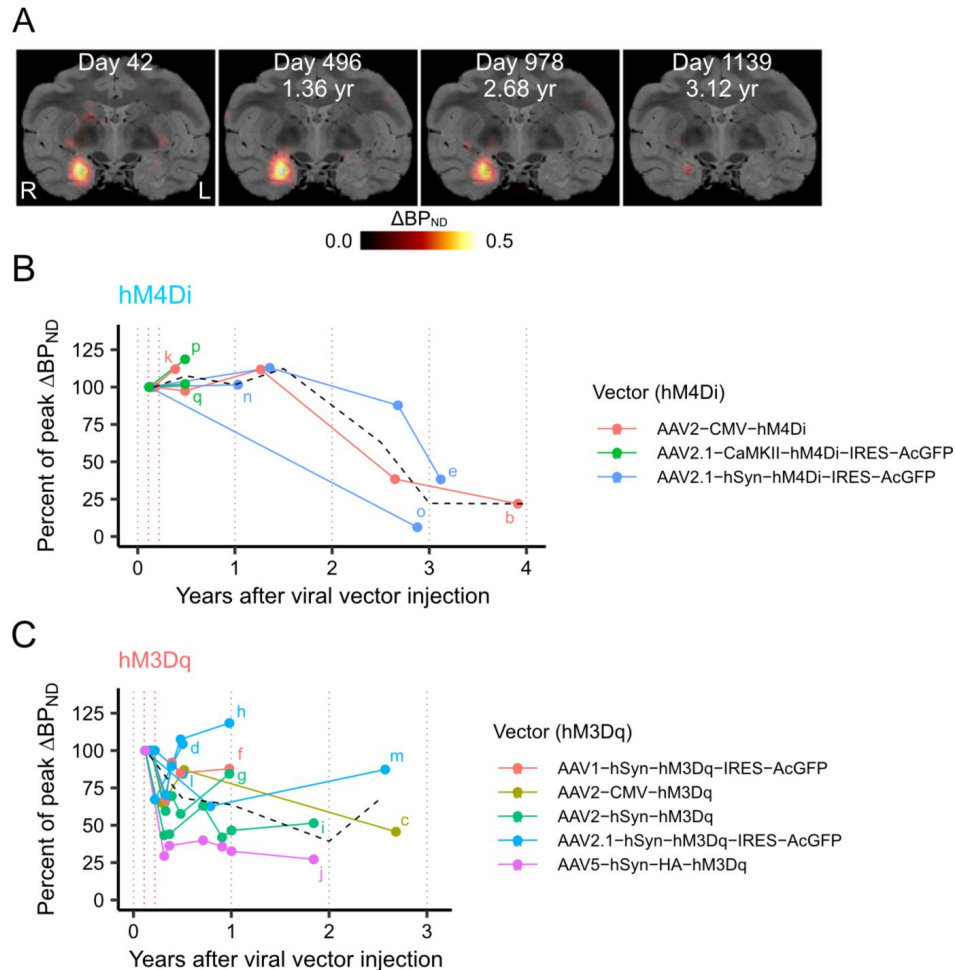


Figure 2

Longitudinal time-course of DREADD expression levels

(A) Parametric coronal images of DREADD expression levels (ΔBP_{ND}) overlaid on MR images obtained from monkey #237, measured at 42, 496, 978, and 1,139 days after injection. (B-C) Time course of hM4Di (B) and hM3Dq (C) expression levels normalized to the peak ΔBP_{ND} observed between 40- and 80-days post-injection (time between the dotted red vertical lines). Each colored line represents one injection site, and the lowercase letters at the ends of each line correspond to the injection sites listed in [Table 2](#). Each dot represents one [^{11}C]DCZ PET measurement. The dashed black line indicates the group average at six-months intervals. hM4Di expression levels remained stable for approximately 1.5 and those for hM3Dq about 1 year, after which variable patterns of decline were observed.

DREADDs effectively modulated neuronal activity and behavior for approximately two years

We assessed data from 17 monkeys to evaluate the duration over which DREADD activation induced by DREADD agonists produced significant effects on neural activity and behavior. Monkeys expressing hM4Di (N = 11) were assessed through behavioral testing, two of which also underwent electrophysiological assessment. Monkeys expressing hM3Dq (N = 6) were assessed for changes in glucose metabolism via [^{18}F]FDG-PET (N = 5) or alterations in neuronal activity using electrophysiology (N = 1). Across these assessments, significant chemogenetic effects were observed for up to three years following AAV vector injection (**Figure 3**). For example, one monkey with bilateral hM4Di expression in the dorsolateral prefrontal cortex (dlPFC; monkey #229) consistently displayed impaired performance on the delayed response task with DCZ administration for up to 2.3 years (867 days) following vector injection (Oyama et al., 2021). Similarly, another monkey with hM3Dq expression in the amygdala (monkey #215) exhibited increased glucose metabolism following DCZ administration for up to 2.5 years (926 days) following vector injection, as measured by FDG-PET (Nagai et al., 2020).

Although experiment durations varied across monkeys, we consistently observed significant effects throughout the study periods, with only one monkey (#234) being removed from experiments due to a loss of DREADD effectiveness three years after transduction. In all other cases, receptor activation was performed repeatedly several tens of times, resulting in no clear loss of efficacy. After the termination of the chemogenetic studies, DREADD expression was verified by *in vitro* via postmortem immunohistochemical analysis (**Fig. 3C-H**). These findings highlight the robust and sustained functionality of DREADD systems in long-term experiments.

As mentioned above, after more than three years, we observed the disappearance of hM4Di expression in one monkey (#234), and this was associated with the loss of chemogenetic behavioral effects. Specifically, activating hM4Di in the functionally defined representation of index-finger primary somatosensory cortex (SI_{D2}) resulted in reversible behavioral deficits for two years, including impaired finger dexterity and hypersensitivity in the foot, confined to the contralateral side (**Figure 4A**) (Hirabayashi et al., 2021). However, *in vivo* imaging more than three years after injection indicated diminished hM4Di expression, and subsequent behavioral assessments failed to detect any motor deficits or hypersensitivity (**Figure 4B**). Post-mortem immunohistochemical examination confirmed the absence hM4Di, with no evidence of neuronal loss at the injection site (**Figure 4C**).

Protein tags reduce peak DREADD expression levels

Finally, we investigated the factors influencing the peak level of DREADD expression. To minimize confounding effects related to injection volume and method, this analysis was limited to cases in which AAV vectors were delivered via microinjector (see Methods).

We applied a linear model incorporating multiple variables, including injection volume, serotype, promoter, titer, tag, and DREADD type to assess how each factor contributed to peak expression levels. Our analysis revealed that the presence and type of co-expressed protein tags significantly affect peak expression levels ($P < 0.008$; **Table 3**). Specifically, peak expression levels were lower for vectors in which the fusion HA-tag sequence was encoded at the 5'-terminal site (5'-HA) of the DREADD sequence than they were for vectors that encoded GFP following the internal ribosome entry site (IRES) sequence, both of which resulted in lower peak expression levels compared with vectors that did not include protein tags (**Figure 5**). A potential interaction between DREADD type and promoter was also observed (**Table 3**; Supplementary Fig. 1); however, given the limited sample size (n = 1 for hM3Dq with CMV), no definitive conclusion can be made.

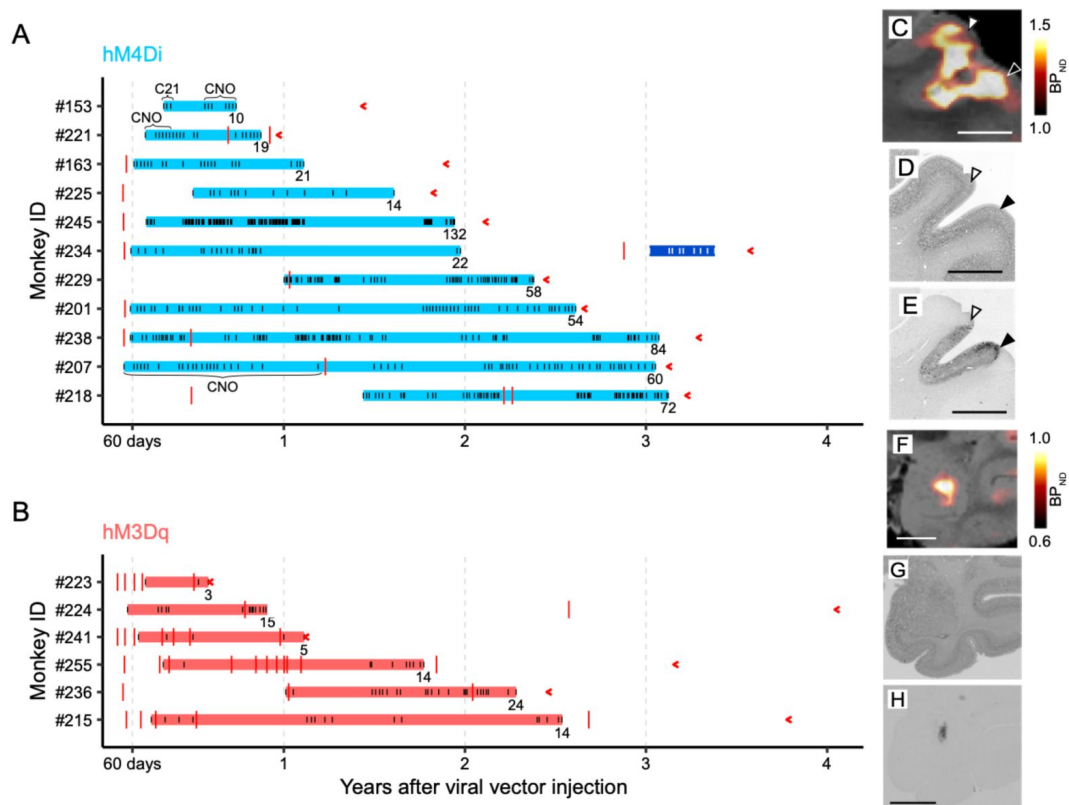


Figure 3

Duration of effective functional modulation induced by DREADDs

(A-B) Duration of effective functional modulation induced by hM4Di (A) and hM3Dq (B). Blue and red horizontal bars indicate the duration of successful chemogenetic manipulation. Black and white ticks on the horizontal bars mark the timing of agonist administration, with CNO and Compound 21 (C21) explicitly stated; all other ticks correspond to DCZ. For example, repeated DCZ administration consistently induced behavioral change in monkey #229 and increased glucose metabolism in monkey #215. The corresponding functional assessments are summarized in [Table 1](#). The dark blue bar indicates the duration of failed chemogenetic manipulation in monkey #234 (see also [Figure 4](#)). Total numbers of agonist administrations (DREADD activation) are shown at the end of each duration bar. Red ticks indicate the timing of DCZ-PET scans and red arrowheads indicate the timing of perfusion. (C) *In vivo* visualization of hM4Di expression in the dorsolateral prefrontal cortex 377 days after injection (monkey #229). The coronal PET image showing specific binding of [¹¹C]DCZ is overlaid on MR images. (D and E) Nissl-(D) and DAB-stained (E) sections corresponding to C, representing immunoreactivity against reporter protein. White and black arrowheads represent the dorsal and ventral borders of the target regions, respectively. Scale bars, 5 mm. (F) *In vivo* visualization of hM3Dq expression in the amygdala 980 days after the injection (monkey #215). The coronal PET image showing specific binding of [¹¹C]DCZ is overlaid on MR images. (G and H) Nissl-(G) and DAB-stained (H) sections corresponding to F, representing immunoreactivity against reporter protein.

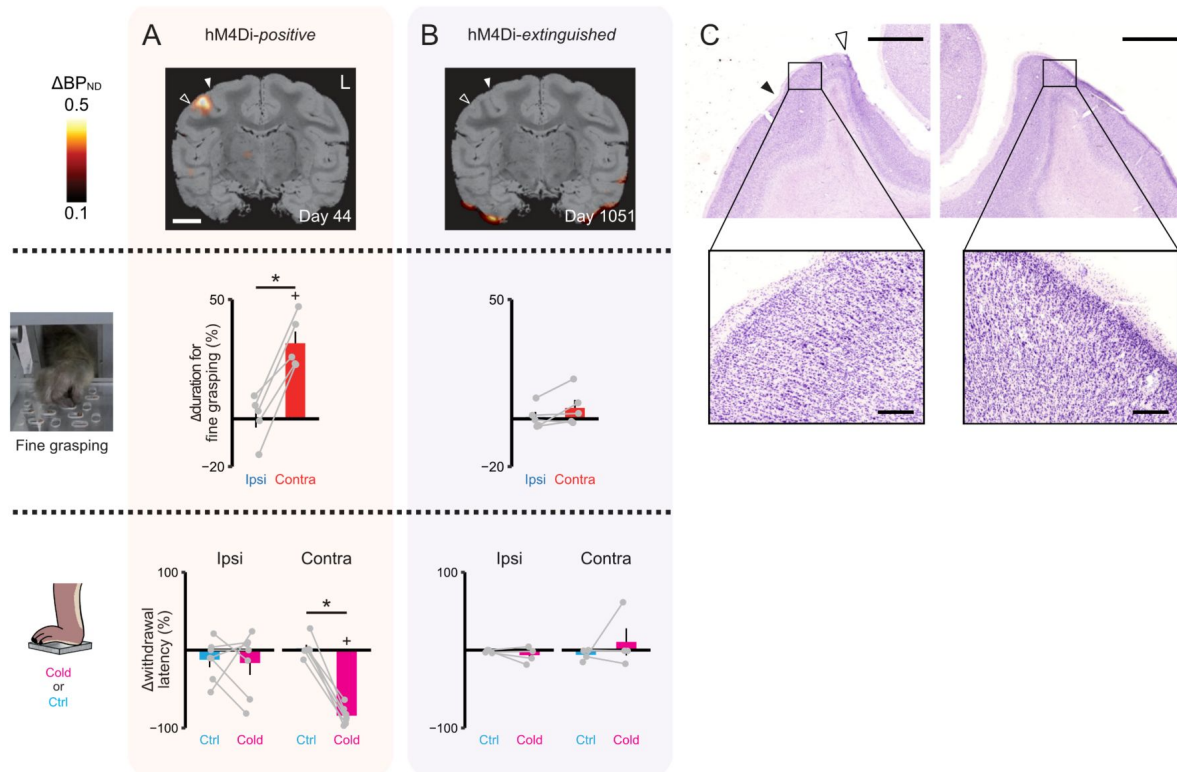


Figure 4.

Disappearance of behavioral effects in an hM4Di-extinguished monkey (#234).

(A) **Top**, Coronal PET image showing ΔBP_{ND} of [^{11}C]DCZ data overlaid on an MR image, obtained 44 days after viral vector injection into the right SI_{D2}. Filled and open arrowheads represent the central and ipsilateral sulci, respectively. **Middle**, Performance of fine grasping using a modified Brinkman board task, assessing the monkey's ability to pick up small food pellets with its thumb and index fingers. Values indicate the change in total duration to complete the task between pre- and post-DCZ administration sessions. *Ipsi* (*Contra*) refers to performance using the hand ipsilateral (contralateral) to the hM4Di-expressing SI_{D2}. Gray lines connect performances from the same sessions using different hands. Error bars, s.e.m. *: $P < 0.002$, paired t-test. +: $P < 0.004$, paired t-test, adjusted for multiple comparisons. $n = 5$ sessions. **Bottom**, DCZ-induced change in the foot withdrawal latency in response to cold (magenta) or control (cyan) stimulation. Negative values indicate faster withdrawal latency following DCZ administration. Gray lines connect performances from the same sessions. Error bars, s.e.m. *: $P < 0.001$, paired t-test. +: $P < 0.001$, paired t-test comparing between pre- and post-DCZ administration. $n = 7$ sessions. (B) Same types of measurements as in A, but following the extinction of hM4Di expression. The PET image was obtained about three years (1,051 days) post-vector injection. $n = 5$ and $n = 4$ sessions for Brinkman board task and foot withdrawal test, respectively. (C) A Nissl-stained section demonstrating the absence of neuronal loss at the vector injection sites (left panel) and the contralateral side (right panel). The locations of the filled and open arrowheads correspond to those shown in A and B. Scale bars are 2.5 mm and 250 μm .

Effect	DFn	DFd	F-value	p-value	effect size (η^2G)
Titer	1	4	2.093	0.222	0.343
DREADD	1	4	2.565	0.184	0.391
Promoter	1	4	5.280	0.083	0.569
Tag	2	4	19.84	0.008*	0.908
Serotype	3	4	2.105	0.242	0.612
Volume	1	4	0.090	0.779	0.022
Titer:Promoter	1	4	2.346	0.200	0.370
DREADD:Promoter	1	4	10.74	0.031*	0.729
Titer:Tag	2	4	1.494	0.328	0.428
Titer:Volume	1	4	3.199	0.148	0.444
DREADD:Volume	1	4	1.497	0.288	0.272

DFn, degrees of freedom numerator; *DFd*, degrees of freedom denominator; η^2G , Generalized eta-squared; *, $p < 0.05$. The ANOVA table describes the factors contributing to the level of expression.

The optimal model, selected based on Akaike's Information Criterion, was the following:

$$\begin{aligned} \Delta BP_{ND} = & \text{viral titer} + \text{DREADD type} + \text{promoter} + \text{reporter tag} + \text{serotype} + \text{injection volume} \\ & + \text{viral titer:promoter} + \text{DREADD type:promoter} + \text{viral titer:reporter-tag} \\ & + \text{viral titer:injection volume} + \text{DREADD type:injection volume}. \end{aligned}$$

The results indicate that the reporter-tag and the interactions between DREADD type:promoter significantly affected DREADD expression levels. However, interpretation of the latter should be made with caution due to limited sampling in some combinations (see Supplementary Fig. 1 and main text).

Table 3.

Result of the linear model analysis

Discussion

DREADDs have emerged as invaluable tools in nonhuman primate research due to their ability to exert effects on non-spatially restricted brain regions, allowing for simultaneous and discrete targeting of multiple areas. This capability makes DREADDs promising for investigating the causal roles of specific neural pathways and cell types in the highly specialized brain circuits found in primates. However, ethical and logistical constraints on using a large number of monkeys highlight the need of moving beyond a trial-and-error approach. It is essential to aggregate data and provide insights that are more widely applicable, thus enabling the development of a more efficient experimental design for DREADD research in monkeys. *In vivo* PET visualization of DREADDs offers a non-invasive and quantitative means for real-time monitoring of receptor expression. The present study analyzed PET and functional/behavioral data from 43 injection sites across 20 monkeys and revealed three key findings: (1) after AAV injection, DREADD expression levels peaked after approximately 60 days and remained stable for up to 1.5 years, followed by a gradual decline observed after two to three years; (2) significant chemogenetic effects on neural activity and behavior persisted for around two years; and (3) protein tags significantly influenced peak expression levels, with co-expressed protein tags reducing peak expression. These results offer quantitative insights into the temporal dynamics of DREADD expression, thereby informing strategies for designing long-term experiments. They also confirm the efficacy and reliability of DREADD-mediated interventions.

Technical considerations

In vivo monitoring of gene expression has been conducted using fluorescent reporters observed through a window or fiber, providing a method to verify transducing protein expression and localization (Diester et al., 2011 [DOI](#); Ruiz et al., 2013 [DOI](#)). Although useful for optogenetics and imaging studies, these methods are invasive, less quantitative, and spatially limited, often requiring cranial surgeries and capturing only two-dimensional data. In contrast, DCZ PET imaging provides a direct, non-invasive, volumetric measure of receptor expression, making it particularly suitable for studies in large animals such as macaques and for translational research with potential application to future human therapies.

Nevertheless, PET measurement has certain limitations, including potential variability due to individual differences, underestimation of DREADD expression levels due to the partial volume effect, and the influence of anesthesia on kinetic parameters during imaging sessions. In this study, we have considered these factors and minimized their impact by applying appropriate ROI placements and excluding data collected under inadequate anesthetic conditions (see Methods).

This study included a retrospective analysis of datasets pooled from multiple studies conducted within a single laboratory, which inherently introduced variability across injection parameters and scan intervals. While such an approach reflects real-world practices in long-term NHP research, future studies, including multicenter efforts using harmonized protocols, will be valuable for systematically assessing inter-individual differences and optimizing key experimental parameters.

Temporal dynamics of DREADD expression

Our [¹¹C]DCZ PET measurement showed that DREADD expression peaked around 60 days after AAV injection, regardless of whether hM4Di or hM3Dq was used. This pattern was consistent with our previous data regarding hM4Di expression visualized using [¹¹C]clozapine (Nagai et al., 2016 [DOI](#)). This gradual rise in AAV-mediated gene expression likely reflects rate-limiting steps following viral infection, including transport to the nucleus, uncoating, and conversion to double-

strand DNA (Duan et al., 2000 [↗](#); Ferrari et al., 1996 [↗](#); Li and Samulski, 2020 [↗](#); Wang et al., 2007 [↗](#)). Similar temporal patterns of AAV-mediated gene expression have been observed in previous studies using *in vivo* fluorescent measurement. For example, in rats transfected with an AAV5 encoding fluorescence protein (EYFP) in motor cortex and the hippocampus, *in vivo* fluorescence signals increased sigmoidally, rapidly until day 35, then slowed down (Diester et al., 2011 [↗](#)). Similarly, in macaques, Chr2 expression monitored via fluorescence intensity peaked at around 60 days after AAV9 injection (Nakamichi et al., 2019 [↗](#)).

Although we modeled the time course of DREADD expression using a single sigmoid function, PET data from several monkeys showed a modest decline following the peak. While the sigmoid model captured the early-phase dynamics and offered a reliable estimate of peak timing, additional PET scans—particularly between 60- and 120-days post-injection—will be essential to fully characterize the biological basis of the post-peak expression trajectories.

Despite being derived from a limited dataset, our findings offer important insights for designing chemogenetics studies in primates. Specifically, beginning functional experiments at approximately 60 days post-AAV injection will yield reliable results and avoid unnecessary delays. This timeline may also be applicable to other genetic studies in monkeys, such as those using optogenetics or calcium imaging, which also require proper expression of functional proteins. However, while these tools rely on rapid, light driven responses and are relatively less dependent on stable functional protein expression, DREADDs require sustained expression for consistent long-term effects.

Factors influencing peak DREADD expression

Although higher AAV titers generally result in increased transgene expression in neurons, excessively high titers can trigger immune responses. In the current study, we used titers that ranged from 1 to 5×10^{13} gc/ml, achieving a balance between effective expression and minimal immune activation. This may explain why virus titer did not emerge as a significant factor affecting expression levels in our analysis. By contrast, our analysis indicated that the presence and type of protein tags significantly influenced peak DREADD expression levels. Vectors encoding DREADDs with protein tags, particularly 5'-terminal HA tags, were associated with a reduction of peak expression levels. Thus, although these tags are commonly used for antibody detection in DREADD constructs, they negatively impact expression efficacy and potentially affect protein function. Our findings are consistent with a previous PET study using [^{11}C]clozapine and immunohistochemical analysis, which also found reduced hM4Di expression in constructs with 5'-terminal HA tags (Nagai et al., 2016 [↗](#)). Additionally, constructs incorporating IRES-GFP might also result in lower expression levels due to the size and composition of the AAV vector, as has been shown in previous studies (Furler et al., 2001 [↗](#); Kimura et al., 2023 [↗](#); Mizuguchi et al., 2000 [↗](#); Zhou et al., 1998 [↗](#)). Nevertheless, these constructs have been frequently used in rodent research and have also been effective in producing long-term chemogenetic effects in monkey studies, including the current one (e.g., monkeys #229 and #238). Thus, while the observed decline in DREADD expression resulting from these constructs is not inherently detrimental to experimental outcomes in monkey studies, it serves as a warning that we must be careful when designing vectors to avoid reductions in experimental efficacy.

Implications for long-term experiments with DREADD expression

One of the most important findings was that AAV-mediated DREADD expression was maintained for up to 2 years for hM4Di and at least 1 year for hM3Dq. Throughout the duration of the experiments, chemogenetic effects remained consistent, as verified by individual behavioral and functional assessments. However, hM4Di expression diminished after 3 years, coinciding with the loss of chemogenetic behavioral effects in at least one case. This indicates that while chemogenetic effects can be maintained for up to two years, they likely decline gradually beyond this period.

Consequently, researchers should anticipate consistent chemogenetic effects for a maximum of two years following AAV injection, and should be mindful that longer-term experiments are at risk for a decline in effectiveness. Specifically, the inevitable failure to detect DREADDs via immunohistochemistry highlights an important caveat for researchers: timely termination of experiments to ensure verification through *in vitro* examination is necessary.

The gradual decline in expression levels observed beyond two years has several potential causes. First, AAV vectors typically remain episomal in transduced cells, without integrating into the host genome (Weitzman and Linden, 2011 [↗](#)). While this poses less of an issue in non-dividing neurons, episomal DNA can still be subject to gradual loss or degradation over time. Second, several mechanisms may contribute the gradual downregulation of transgene expression, including epigenetic modifications, immune responses, and genome stability (Muhuri et al., 2022 [↗](#)). For example promoter silencing—often associated with epigenetic modifications—has been reported (Gray et al., 2011 [↗](#)). Neuron-specific promoters have been shown to achieve more efficient and prolonged transgene expression than ubiquitous promoters like CMV (Paterna et al., 2000 [↗](#)).

Our study found no evidence of cell death at AAV injection sites (Figure 4C [↗](#)), suggesting that neuronal loss was not a primary cause of the eventual decline in expression levels. Repeated receptor activation did not affect DREADD expression levels or chemogenetic efficacy in a previous study (Oyama et al., 2022). Thus, identifying the factors that cause expression to eventually decline remains of critical importance. With this knowledge we can develop strategies to maintain persistent transgene expression in neurons for more than two years, which will benefit both chemogenetic studies and therapeutic applications. Despite this uncertainty, our data offers important guidelines for chemogenetics studies in nonhuman primates, demonstrating that up to two years of reliable expression is achievable.

Translational implications

This research has significant translational implications, particularly for the development of DREADD-based therapies for treating neurological and psychiatric disorders. DREADD-based approaches have shown promising therapeutic potential, with successful application in monkey models of disease conditions such as epileptiform seizures (Miyakawa et al., 2023 [↗](#)) and Parkinson's disease (Chen et al., 2023 [↗](#)). Extending these findings to human clinical applications positions PET imaging as a powerful tool for real-time, non-invasive confirmation of target protein expression—a critical factor for ensuring therapeutic efficacy. PET reporter imaging has already shown its value in human gene therapy for certain movement disorders, where sustained protein expression has been correlated with treatment efficacy (Mittermeyer et al., 2012 [↗](#); Tai et al., 2022 [↗](#)). Our study reinforces this utility by demonstrating that when DREADD expression was no longer detectable via PET, the chemogenetic effects were also diminished.

Conclusion

Our study describes short- and long-term dynamics of DREADD expression in nonhuman primates using [¹¹C]DCZ PET imaging, complemented by assessment of functional efficacy. The findings provide critical insights into the practicality of multi-year studies using chemogenetic neuromodulation with DREADDs, including the expression time course and factors affecting expression stability. This work offers valuable guidance for designing future long-term nonhuman primate studies, and underscores the translational potential of using DREADDs combined with PET imaging for non-invasive, gene-targeted therapies for neurological and psychiatric disorders.

Methods

Subjects

A total of 20 macaque monkeys (10 Japanese, *Macaca fuscata*; 8 Rhesus, *Macaca mulatta*; 2 Cynomolgus, *Macaca fascicularis*; 16 males, 4 females; weight, 3.4–9.9 kg; age, 3–13 years at the beginning of experiments) were used (**Table 1**). The monkeys were kept in individual primate cages in an air-conditioned room. A standard diet, supplementary fruits/vegetables, and a tablet of vitamin C (200 mg) were provided daily. All experimental procedures involving animals were carried out in accordance with the Guide for the Care and Use of Nonhuman Primates in Neuroscience Research (The Japan Neuroscience Society; https://www.jnss.org/en/animal_primates) and were approved by the Animal Ethics Committee of the National Institutes for Quantum Science and Technology (11-1038).

Viral vector production

We used a total of 11 adeno-associated virus (AAV) vectors. AAV2-Syn-HA-hM3Dq, AAV2-Syn-HA-hM4Di, and AAV5-Syn-HA-hM4Di were purchased from Addgene (MA, USA). The other viral vectors were produced by a helper-free triple transfection procedure, which was purified by affinity chromatography (GE Healthcare, Chicago, USA). Viral titer was determined by quantitative polymerase chain reaction using Taq-Man technology (Life Technologies, Waltham, USA).

Surgical procedures and viral vector injections

Surgical procedures have been described in detail in previous reports (Hirabayashi et al., 2021; Nagai et al., 2020, 2016; Oyama et al., 2023, 2022a, 2021). The AAV vectors and brain regions into which they were injected are summarized in **Table 2**. Before surgery, structural scans of the head were acquired via magnetic resonance (MR) imaging (7 tesla 400 mm/SS system, Bruker, Billerica, MA, USA) and X-ray computed tomography (CT) (Accutomo170, J. MORITA CO., Kyoto, Japan) under anesthesia (continuous intravenous infusion of propofol 0.2–0.6 mg/kg/min). Overlay MR and CT images were created using PMOD® image analysis software (PMOD 4.4, PMOD Technologies Ltd, Zurich, Switzerland) to estimate the stereotaxic coordinates of target brain structures.

Surgeries were performed under aseptic conditions in a fully equipped operating suite. We monitored body temperature, heart rate, peripheral oxygen saturation (SpO₂), and endo-tidal CO₂ throughout all surgical procedures. Monkeys were immobilized by intramuscular (i.m.) injection of ketamine (5–10 mg/kg) and xylazine (0.2–0.5 mg/kg) after i.m. injection of atropine sulfate (0.02–0.05 mg/kg) and then intubated with an endotracheal tube. Anesthesia was maintained with isoflurane (1%–3%, to effect). After surgery, prophylactic antibiotics and analgesics (cefmetazole, 25–50 mg/kg/day; ketoprofen, 1–2 mg/kg/day) were administered for 7 days.

Viral vectors were injected with a stereotaxic apparatus or a hand-held syringe. For stereotaxic injection, burr holes (~8 mm in diameter) were created for inserting the injection needle. Viruses were pressure-injected using a 10-μl Hamilton syringe (model 1701 RN, Hamilton, Reno, NV) mounted on a motorized microinjector (UMP3T-2, WPI, Sarasota, FL) held by a manipulator (model 1460, David Kopf, Tujunga, CA) on the stereotaxic frame. After making an approximately 3-mm cut in the dura mater, the injection needle was inserted into the brain and slowly moved down 1 to 2 mm beyond the target and then kept stationary for 5 min, after which it was pulled back up to the target location. Injection speed was 0.1 to 0.5 μl/min. After each injection, the needle remained *in situ* for 15 min to minimize backflow along the needle and then was pulled out very slowly.

For handheld injection, the target cortex was exposed by removing a bone flap and reflecting the dura mater. Injections were made under visual guidance through an operating microscope (Leica M220, Leica Microsystems GmbH, Wetzlar, Germany) at an oblique angle to the brain surface.

Each hemisphere received nine tracks of injections: one at the caudal tip, four along the dorsal bank, and four along the ventral bank of the principal sulcus, with 3 to 5 μL per track, depending on the depth (Oyama et al., 2023 [DOI](#)).

PET imaging

PET imaging was conducted as previously reported (Nagai et al., 2020 [DOI](#)). Briefly, PET scans were performed using a microPET Focus 220 scanner (Siemens Medical Solutions USA, Malvern, USA). Monkeys were initially immobilized with ketamine (5–10 mg/kg) and xylazine (0.2–0.5 mg/kg) and maintained under anesthesia with isoflurane (1%–3%) throughout the PET procedures. The isoflurane level was carefully adjusted to ensure the stability of continuously monitoring physiological parameters, such as body temperature, heart rate, SpO_2 , and endo-tidal CO_2 . Endo-tidal CO_2 levels were maintained between 35 and 45 mmHg to prevent substantial changes in cerebral blood flow (Markwalder et al., 1984 [DOI](#)), which could impact tracer kinetics. Transmission scans were performed for about 20 min with a Ge-68 source. Emission scans were acquired in 3D list mode with an energy window of 350–750 keV after intravenous bolus injection of [^{11}C]DCZ (~350 MBq) and [^{18}F]FDG (~200 MBq). Data acquisition lasted for 90 min.

To estimate the specific binding of [^{11}C]DCZ, regional binding potential relative to a non-displaceable radioligand (BP_{ND}) was calculated by PMOD[®] with an original multilinear reference tissue model (MRTMo) (Yan et al., 2021 [DOI](#)). The cerebellum was used as a reference region, and t^* was 15 min. The DREADD expression level was defined as $\Delta\text{BP}_{\text{ND}}$, calculated by subtracting the pre-injection value from the post-injection value because the BP_{ND} value included off-target binding. This minimized individual differences because off-target binding varied among subjects. The volumes of interest (VOIs) reflecting DREADD expression were identified by drawing contours at half of the maximum $\Delta\text{BP}_{\text{ND}}$ value in the $\Delta\text{BP}_{\text{ND}}$ parametric image, which was smoothed with a 2-mm Gaussian filter, and overlaid onto the corresponding region in each individual MR image.

For the FDG study, DCZ (1 $\mu\text{g/kg}$) or vehicle was administered intravenously 1 min before FDG injection. Data were converted to Standardized Uptake Value (SUV) images, averaged between 30- and 60-min, and normalized to SUV ratio (SUVR) images using the mean whole-brain value. Metabolic changes induced by DCZ were calculated as ΔSUVR and defined by the different DCZ and vehicle conditions, with VOIs for FDG analysis matching those used in the [^{11}C]DCZ-PET study.

Administration of DREADD agonists

Deschloroclozapine (DCZ; HY-42110, MedChemExpress) was the primary agonist used. DCZ was first dissolved in dimethyl sulfoxide (DMSO; FUJIFILM Wako Pure Chemical Corp.) and then diluted in saline to a final volume of 1 mL, with the final DMSO concentration adjusted to 2.5% or less. DCZ was administered intramuscularly at a dose of 0.1 mg/kg for hM4Di activation, and at 1–3 $\mu\text{g/kg}$ for hM3Dq activation. For behavioral testing, DCZ was injected approximately 15 min before the start of the experiment unless otherwise noted. Fresh DCZ solutions were prepared daily.

In a limited number of cases, clozapine-N-oxide (CNO; Toronto Research Chemicals) or Compound 21 (C21; Tocris) was used as an alternative DREADD agonist for some hM4Di experiments. Both compounds were dissolved in DMSO and then diluted in saline to a final volume of 2–3 mL, also maintaining DMSO concentrations below 2.5%. CNO and C21 were administered intravenously at doses of 3 mg/kg and 0.3 mg/kg, respectively.

Behavioral tasks

The behavioral experiments are summarized in **Table 1** [DOI](#). Complete descriptions of the detailed procedures for each behavioral experiment are available in previous reports (Hirabayashi et al., 2024 [DOI](#), 2021 [DOI](#); Hori et al., 2021 [DOI](#); Oyama et al., 2024 [DOI](#), 2022b [DOI](#), 2021 [DOI](#)). Here we focus on a modified Brinkman board task and a sensitivity to cold stimulation task, which were used to

analyze the behavior of a monkey (#234) that had hM4Di expressed in its primary somatosensory cortex (**Fig. 4**). In the modified Brinkman board task, monkeys picked food pellets out of 10 slots in a board with one of their hands (i.e., contralateral or ipsilateral to the DREADD-expressing hemisphere), and the amount of time needed to pick up all ten pellets was recorded. The time to complete each trial was normalized to the number of pellets successfully picked up to obtain a value that reflected manual dexterity. In a single experimental session, a monkey performed five to ten trials with each hand before and 10 min after intravenous systemic administration of DCZ. In the sensitivity to cold task, monkeys were trained to place a sole of one of their feet on a cold metal plate that was placed in front of their monkey chair. The latency with which they withdrew their foot from the plate was recorded and used as an index of sensitivity to a cutaneous (cold) stimulus. If a monkey did not withdraw its foot within 60 s, the trial was terminated and a withdraw latency of 60 s was assigned for that trial. The monkeys performed five to ten trials for each of two temperature conditions (30°C for control and 10°C for cold stimulation) both before and 10 min after DCZ administration. Chemogenetic effects were assessed by comparing performance on the tasks pre- and post-injection and between contralateral and ipsilateral hands and feet (only the contralateral side should be affected by DCZ).

Data analysis and statistics

All data and statistical analyses were performed using the R statistical computing environment (R Development Core Team, 2004). To model the time course of DREADD expression, we used a single sigmoid function, referencing past *in vivo* fluorescent measurements (Diester et al., 2011). Curve fitting was performed using least squares minimization. For comparison, a double logistic function was also tested and evaluated using the Bayesian Information Criterion (BIC) to assess model fit. Repeated-measures two-way ANOVAs were used to examine the effect of cumulative number of activation times $\times$ days after injection on post-peak fluctuation.

To identify factors influencing peak expression levels, a stepwise model selection using the stepAIC() function from the MASS package (Venables and Ripley, 2002) was performed. To minimize potentially confounding effects related to injection method and volume, we restricted this analysis to cases in which a microinjector was used to deliver the AAV, and excluded cortical injections into the orbitofrontal cortex and dorsolateral prefrontal cortex, which involved larger, manually administered injection volumes. Starting with a full model that included six factors—virus titer, DREADD type, promoter, tag, serotype, and injection volume—along with all their interactions, the best-fitting linear model was determined based on Akaike information criterion (AIC). A permutation-based outlier analysis was also performed to detect and exclude statistical outliers that could disproportionally influence the regression results. The resulting best model included six variables and five interactions (Titer:Promoter, DREADD:Promoter, Titer:Tag, Titer:Volume, and DREADD:Volume). The significance of their contributions to the peak expression level, as well as the effect sizes (generalized eta-squared, η^2G), were assessed using Type II analysis of variance with the anova_test() function from the rstatix package (Kassambara, 2023).

Histology and immunostaining

For histological inspection, monkeys were deeply anesthetized with sodium pentobarbital (50 mg/kg, i.v.) or sodium thiopental (50 mg/kg, i.v.) after administration of ketamine hydrochloride (5–10 mg/kg, i.m.) and then transcardially perfused with saline at 4°C, followed by 4% paraformaldehyde in 0.1 M phosphate buffered saline (PBS, pH 7.4) at 4°C. The brain was removed from the skull, postfixed in the same fresh fixative overnight, saturated with 30% sucrose in phosphate buffer (PB) at 4°C, and then cut serially into 50- μ m-thick sections on a freezing microtome. For visualizing immunoreactive GFP (co-expressed with hM4Di) signals, a series of every 6th section was immersed in 1% skim milk for 1 h at room temperature and incubated overnight at 4°C with rabbit anti-GFP monoclonal antibody (1:500, G10362, Thermo Fisher Scientific, Waltham, MA, USA) in PBS containing 0.1% Triton X-100 and 1% normal goat serum for 2 days at 4°C. The sections were then incubated in the same fresh medium containing biotinylated

goat anti-rabbit IgG antibody (1:1,000; Jackson ImmunoResearch, West Grove, PA, USA) for 2 h at room temperature, followed by avidin-biotin-peroxidase complex (ABC Elite, Vector Laboratories, Burlingame, CA, USA) for 2 h at room temperature. For visualizing the antigen, the sections were reacted in 0.05 M Tris-HCl buffer (pH 7.6) containing 0.04% diaminobenzidine (DAB), 0.04% NiCl_2 , and 0.003% H_2O_2 . The sections were mounted on gelatin-coated glass slides, air-dried, and cover-slipped. Parts of other sections were Nissl-stained with 1% Cresyl violet. Images of sections were digitally captured using an optical microscope equipped with a high-grade charge-coupled device (CCD) camera (Biorevo, Keyence, Osaka, Japan).

Data Availability

Source data to reproduce the main results of the paper presented in all figures are provided at an open-source repository (<https://github.com/minamimoto-lab/2024-Nagai-LongitudinalPET.git> )

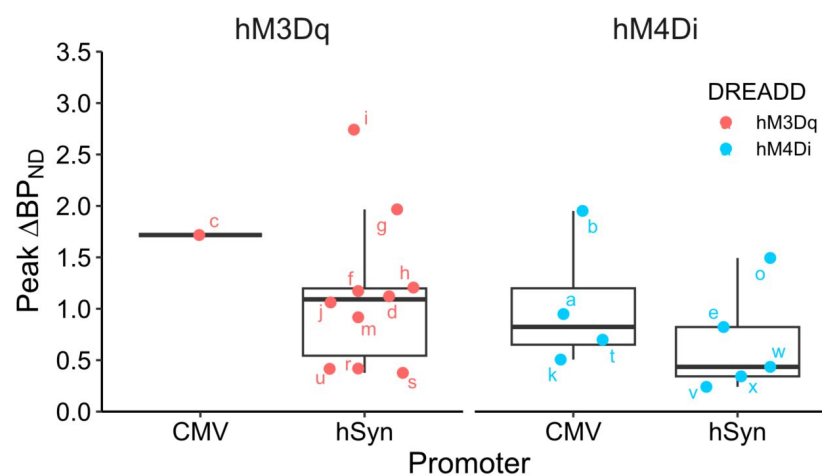
Acknowledgements

This study was supported by MEXT/JSPS KAKENHI Grant Numbers JP19K08138, JP23K27098 (to YN), 24H00734 (to TH), 22H05157 (to KI), and JP20H05955 (to TM), by JST PRESTO Grant Numbers JPMJPR22S3 (to KO), by AMED Grant Numbers JP23wm0625001 (to TM) and JP24wm0625307 (to TH), and by the Moonshot Research & Development Program (Millennia Program) from JST Grant Number JPMJMS2295 (to TM and KI). Japanese monkeys were provided by the Japan MEXT National Bio-Resource Project “Japanese Monkeys”. We thank Jun Kamei, Ryuji Yamaguchi, Yuichi Matsuda, Yoshio Sugii, Takashi Okauchi, Risa Suma, Tomomi Kokufuta, Rie Yoshida, and Akari Ueshiba for their technical assistance.

Additional information

Author contributions

Conceptualization: YN and TM; Resources: KI and KK; Investigation: YN, Yukiko H, TH, KO, and NM; Visualization: YN, KM, and TH; Formal analysis: YN, KM, and TH; Writing—original draft: YN, TH, and TM; Writing—review & editing: All authors.



Supplementary Figure 1.

Effect of promoter and DREADD type on peak expression levels.

Peak expression levels (ΔBP_{ND}) are shown for different constructs combining two promoter types and two DREADD types. Box plots represent the median (central line), interquartile range (box), and the range (whiskers) of the data. Individual data points labeled with lowercase letters correspond to the injection sites listed in [Table 2](#).

References

1. Bonaventura J, Eldridge MAG, Hu F, Gomez JL, Sanchez-Soto M, Abramyan AM, Lam S, Boehm MA, Ruiz C, Farrell MR, Moreno A, Galal Faress IM, Andersen N, Lin JY, Moaddel R, Morris PJ, Shi L, Sibley DR, Mahler SV, Nabavi S, Pomper MG, Bonci A, Horti AG, Richmond BJ, Michaelides M (2019) **High-potency ligands for DREADD imaging and activation in rodents and monkeys** *Nat Commun* **10**:4627 [Google Scholar](#)
2. Chen Y, Hong Z, Wang J, Liu K, Liu J, Lin J, Feng S, Zhang T, Shan L, Liu T, Guo P, Lin Y, Li T, Chen Q, Jiang X, Li A, Li X, Li Y, Wilde JJ, Bao J, Dai J, Lu Z (2023) **Circuit-specific gene therapy reverses core symptoms in a primate Parkinson's disease model** *Cell* **186**:5394–5410 [Google Scholar](#)
3. Diester I, Kaufman MT, Mogri M, Pashaie R, Goo W, Yizhar O, Ramakrishnan C, Deisseroth K, Shenoy KV (2011) **An optogenetic toolbox designed for primates** *Nat Neurosci* **14**:387–397 [Google Scholar](#)
4. Duan D, Yue Y, Yan Z, Yang J, Engelhardt JF (2000) **Endosomal processing limits gene transfer to polarized airway epithelia by adeno-associated virus** *J Clin Invest* **105**:1573–1587 [Google Scholar](#)
5. Ferrari FK, Samulski T, Shenk T, Samulski RJ (1996) **Second-strand synthesis is a rate-limiting step for efficient transduction by recombinant adeno-associated virus vectors** *J Virol* **70**:3227–3234 [Google Scholar](#)
6. Furler S, Paterna JC, Weibel M, Büeler H (2001) **Recombinant AAV vectors containing the foot and mouth disease virus 2A sequence confer efficient bicistronic gene expression in cultured cells and rat substantia nigra neurons** *Gene Ther* **8**:864–873 [Google Scholar](#)
7. Gray SJ, Foti SB, Schwartz JW, Bachaboina L, Taylor-Blake B, Coleman J, Ehlers MD, Zylka MJ, McCown TJ, Samulski RJ (2011) **Optimizing promoters for recombinant adeno-associated virus-mediated gene expression in the peripheral and central nervous system using self-complementary vectors** *Hum Gene Ther* **22**:1143–1153 [Google Scholar](#)
8. Grayson DS, Bliss-Moreau E, Machado CJ, Bennett J, Shen K, Grant KA, Fair DA, Amaral DG (2016) **The rhesus monkey connectome predicts disrupted functional networks resulting from pharmacogenetic inactivation of the amygdala** *Neuron* **91**:453–466 [Google Scholar](#)
9. Hirabayashi T, Nagai Y, Hori Y, Inoue K-I, Aoki I, Takada M, Suhara T, Higuchi M, Minamimoto T (2021) **Chemogenetic sensory fMRI reveals behaviorally relevant bidirectional changes in primate somatosensory network** *Neuron* **109**:3312–3322 [Google Scholar](#)
10. Hirabayashi T, Nagai Y, Yuki Hori, Yukiko Hori, Oyama K, Mimura K, Miyakawa N, Iwaoki H, Inoue K-I, Suhara T, Takada M, Higuchi M, Minamimoto T (2024) **Multiscale chemogenetic dissection of fronto-temporal top-down regulation for object memory in primates** *Nat Commun* **15**:5369 [Google Scholar](#)
11. Hori Y, Mimura K, Nagai Y, Fujimoto A, Oyama K, Kikuchi E, Inoue K-I, Takada M, Suhara T, Richmond BJ, Minamimoto T (2021) **Single caudate neurons encode temporally discounted**

12. Ji B, Kaneko H, Minamimoto T, Inoue H, Takeuchi H, Kumata K, Zhang M-R, Aoki I, Seki C, Ono M, Tokunaga M, Tsukamoto S, Tanabe K, Shin R-M, Minamihisamatsu T, Kito S, Richmond BJ, Suhara T, Higuchi M (2016) **Multimodal imaging for DREADD-expressing neurons in living brain and their application to implantation of iPSC-derived neural progenitors** *J Neurosci* 36:11544–11558 [Google Scholar](#)
13. Kassambara A (2023) **rstatix: Pipe-Friendly Framework for Basic Statistical Tests** *CRAN* <https://doi.org/10.32614/CRAN.package.rstatix>
14. Kimura K, Nagai Y, Hatanaka G, Fang Y, Tanabe S, Zheng A, Fujiwara M, Nakano M, Hori Y, Takeuchi RF, Inagaki M, Minamimoto T, Fujita I, Inoue K-I, Takada M (2023) **A mosaic adeno-associated virus vector as a versatile tool that exhibits high levels of transgene expression and neuron specificity in primate brain** *Nat Commun* 14:4762 [Google Scholar](#)
15. Li C, Samulski RJ (2020) **Engineering adeno-associated virus vectors for gene therapy** *Nat Rev Genet* 21:255–272 [Google Scholar](#)
16. Markwalder TM, Grolimund P, Seiler RW, Roth F, Aaslid R (1984) **Dependency of blood flow velocity in the middle cerebral artery on end-tidal carbon dioxide partial pressure--a transcranial ultrasound Doppler study** *J Cereb Blood Flow Metab* 4:368–372 [Google Scholar](#)
17. Mimura K, Nagai Y, Inoue K-I, Matsumoto J, Hori Y, Sato C, Kimura K, Okauchi T, Hirabayashi T, Nishijo H, Yahata N, Takada M, Suhara T, Higuchi M, Minamimoto T. (2021) **Chemogenetic activation of nigrostriatal dopamine neurons in freely moving common marmosets** *iScience* 24:103066 [Google Scholar](#)
18. Mittermeyer G, Christine CW, Rosenbluth KH, Baker SL, Starr P, Larson P, Kaplan PL, Forsayeth J, Aminoff MJ, Bankiewicz KS (2012) **Long-term evaluation of a phase 1 study of AADC gene therapy for Parkinson's disease** *Hum Gene Ther* 23:377–381 [Google Scholar](#)
19. Miyakawa N, Nagai Y, Hori Y, Mimura K, Orihara A, Oyama K, Matsuo T, Inoue K-I, Suzuki T, Hirabayashi T, Suhara T, Takada M, Higuchi M, Kawasaki K, Minamimoto T. (2023) **Chemogenetic attenuation of cortical seizures in nonhuman primates** *Nat Commun* 14:971 [Google Scholar](#)
20. Miyake T, Tanaka K, Inoue Y, Nagai Y, Nishimura R, Seta T, Nakagawa S, Inoue K-I, Hasegawa E, Minamimoto T, Doi M. (2024) **Size-reduced DREADD derivatives for AAV-assisted multimodal chemogenetic control of neuronal activity and behavior** *Cell Rep Methods* 4:100881 [Google Scholar](#)
21. Mizuguchi H, Xu Z, Ishii-Watabe A, Uchida E, Hayakawa T (2000) **IRES-dependent second gene expression is significantly lower than cap-dependent first gene expression in a bicistronic vector** *Mol Ther* 1:376–382 [Google Scholar](#)
22. Mueller SAL, Oler JA, Roseboom PH, Aggarwal N, Kenwood MM, Riedel MK, Elam VR, Olsen ME, DiFilippo AH, Christian BT, Hu X, Galvan A, Boehm MA, Michaelides M, Kalin NH (2023) **DREADD-mediated amygdala activation is sufficient to induce anxiety-like responses in young nonhuman primates** *Curr Res Neurobiol* 5:100111 [Google Scholar](#)
23. Muhuri M, Levy DI, Schulz M, McCarty D, Gao G (2022) **Durability of transgene expression after rAAV gene therapy** *Mol Ther* 30:1364–1380 [Google Scholar](#)

24. Nagai Y, Kikuchi E, Lerchner W, Inoue K-I, Ji B, Eldridge MAG, Kaneko H, Kimura Y, Oh-Nishi A, Hori Y, Kato Y, Hirabayashi T, Fujimoto A, Kumata K, Zhang M-R, Aoki I, Suhara T, Higuchi M, Takada M, Richmond BJ, Minamimoto T (2016) **PET imaging-guided chemogenetic silencing reveals a critical role of primate rostromedial caudate in reward evaluation** *Nat Commun* **7**:13605 [Google Scholar](#)
25. Nagai Y, Miyakawa N, Takuwa H, Hori Y, Oyama K, Ji B, Takahashi M, Huang X-P, Slocum ST, DiBerto JF, Xiong Y, Urushihata T, Hirabayashi T, Fujimoto A, Mimura K, English JG, Liu J, Inoue K-I, Kumata K, Seki C, Ono M, Shimojo M, Zhang M-R, Tomita Y, Nakahara J, Suhara T, Takada M, Higuchi M, Jin J, Roth BL, Minamimoto T (2020) **Deschloroclozapine, a potent and selective chemogenetic actuator enables rapid neuronal and behavioral modulations in mice and monkeys** *Nat Neurosci* **23**:1157–1167 [Google Scholar](#)
26. Nakamichi Y, Okubo K, Sato T, Hashimoto M, Tanifuji M (2019) **Optical intrinsic signal imaging with optogenetics reveals functional cortico-cortical connectivity at the columnar level in living macaques** *Sci Rep* **9**:6466 [Google Scholar](#)
27. Nerella SG, Michaelides M, Minamimoto T, Innis RB, Pike VW, Eldridge MAG (2023) **PET reporter systems for the brain** *Trends Neurosci* **46**:941–952 [Google Scholar](#)
28. Oyama K, Hori Y, Nagai Y, Miyakawa N, Mimura K, Hirabayashi T, Inoue K-I, Suhara T, Takada M, Higuchi M, Minamimoto T (2021) **Chemogenetic dissection of the primate prefronto-subcortical pathways for working memory and decision-making** *Sci Adv* **7**:eabg4246 [Google Scholar](#)
29. Oyama K, Hori Y, Nagai Y, Miyakawa N, Mimura K, Hirabayashi T, Inoue K-I, Takada M, Higuchi M, Minamimoto T (2022a) **Chronic behavioral manipulation via orally delivered chemogenetic actuator in macaques** *J Neurosci* **42**:2552–2561 [Google Scholar](#)
30. Oyama K, Yukiko Hori, Mimura K, Nagai Y, Eldridge MAG, Saunders RC, Miyakawa N, Hirabayashi T, Yuki Hori, Inoue K-I, Suhara T, Takada M, Higuchi M, Richmond BJ, Minamimoto T (2022b) **Chemogenetic disconnection between the orbitofrontal cortex and the rostromedial caudate nucleus disrupts motivational control of goal-directed action** *J Neurosci* **42**:6267–6275 [Google Scholar](#)
31. Oyama K, Majima K, Nagai Y, Yukiko Hori, Hirabayashi T, Eldridge MAG, Mimura K, Miyakawa N, Fujimoto A, Yuki Hori, Iwaoki H, Inoue K-I, Saunders RC, Takada M, Yahata N, Higuchi M, Richmond BJ, Minamimoto T (2024) **Distinct roles of monkey OFC-subcortical pathways in adaptive behavior** *Nat Commun* **15**:6487 [Google Scholar](#)
32. Oyama K, Nagai Y, Minamimoto T (2023) **Targeted delivery of chemogenetic adeno-associated viral vectors to cortical sulcus regions in macaque monkeys by handheld injections** *Bio Protoc* **13**:e4897 [Google Scholar](#)
33. Paterna JC, Moccetti T, Mura A, Feldon J, Büeler H (2000) **Influence of promoter and WHV post-transcriptional regulatory element on AAV-mediated transgene expression in the rat brain** *Gene Ther* **7**:1304–1311 [Google Scholar](#)
34. Roseboom PH, Mueller SAL, Oler JA, Fox AS, Riedel MK, Elam VR, Olsen ME, Gomez JL, Boehm MA, DiFilippo AH, Christian BT, Michaelides M, Kalin NH (2021) **Evidence in primates supporting the use of chemogenetics for the treatment of human refractory neuropsychiatric disorders** *Mol Ther* **29**:3484–3497 [Google Scholar](#)
35. Roth BL (2016) **DREADDs for neuroscientists** *Neuron* **89**:683–694 [Google Scholar](#)

36. Ruiz O, Lustig BR, Nassi JJ, Cetin A, Reynolds JH, Albright TD, Callaway EM, Stoner GR, Roe AW (2013) **Optogenetics through windows on the brain in the nonhuman primate** *J Neurophysiol* **110**:1455–1467 [Google Scholar](#)
37. Tai C-H, Lee N-C, Chien Y-H, Byrne BJ, Muramatsu S-I, Tseng S-H, Hwu W-L (2022) **Long-term efficacy and safety of eladocogene exuparvovec in patients with AADC deficiency** *Mol Ther* **30**:509–518 [Google Scholar](#)
38. Upright NA, Brookshire SW, Schnebelen W, Damatac CG, Hof PR, Browning PGF, Croxson PL, Rudebeck PH, Baxter MG (2018) **Behavioral effect of chemogenetic inhibition is directly related to receptor transduction levels in rhesus monkeys** *J Neurosci* **38**:7969–7975 [Google Scholar](#)
39. Vandeputte C, Evens N, Toelen J, Deroose CM, Bosier B, Ibrahim A, Van der Perren A, Gijsbers R, Janssen P, Lambert DM, Verbruggen A, Debyser Z, Bormans G, Baekelandt V, Van Laere K. (2011) **A PET brain reporter gene system based on type 2 cannabinoid receptors** *J Nucl Med* **52**:1102–1109 [Google Scholar](#)
40. Venables WN, Ripley BD (2002) **Modern Applied Statistics with S** New York: Springer [Google Scholar](#)
41. Wang J, Xie J, Lu H, Chen L, Hauck B, Samulski RJ, Xiao W (2007) **Existence of transient functional double-stranded DNA intermediates during recombinant AAV transduction** *Proc Natl Acad Sci U S A* **104**:13104–13109 [Google Scholar](#)
42. Weitzman MD, Linden RM (2011) **Adeno-associated virus biology** *Methods Mol Biol* **807**:1–23 [Google Scholar](#)
43. Yan X, Telu S, Dick RM, Liow J-S, Zanotti-Fregonara P, Morse CL, Manly LS, Gladding RL, Shrestha S, Lerchner W, Nagai Y, Minamimoto T, Zoghbi SS, Innis RB, Pike VW, Richmond BJ, Eldridge MA. (2021) **[11C]deschloroclozapine is an improved PET radioligand for quantifying a human muscarinic DREADD expressed in monkey brain** *J Cereb Blood Flow Metab* **41**:2571–2582 [Google Scholar](#)
44. Yoon SY, Gay-Antaki C, Ponde DE, Poptani H, Vite CH, Wolfe JH (2014) **Quantitative, noninvasive, in vivo longitudinal monitoring of gene expression in the brain by co-AAV transduction with a PET reporter gene** *Mol Ther Methods Clin Dev* **1**:14016 [Google Scholar](#)
45. Zhou Y, Aran J, Gottesman MM, Pastan I (1998) **Co-expression of human adenosine deaminase and multidrug resistance using a bicistronic retroviral vector** *Hum Gene Ther* **9**:287–293 [Google Scholar](#)

Author information

Yuji Nagai

Advanced Neuroimaging Center, National Institutes for Quantum Science and Technology, Chiba, Japan
 ORCID iD: [0000-0001-7005-0749](https://orcid.org/0000-0001-7005-0749)

For correspondence: nagai.yuji@qst.go.jp

Yukiko Hori

Advanced Neuroimaging Center, National Institutes for Quantum Science and Technology,
Chiba, Japan

Ken-ichi Inoue

System Neuroscience Section, Center for the Evolutionary Origins of Human Behavior, Kyoto
University, Inuyama, Japan
ORCID iD: [0000-0002-5683-5280](https://orcid.org/0000-0002-5683-5280)

Toshiyuki Hirabayashi

Advanced Neuroimaging Center, National Institutes for Quantum Science and Technology,
Chiba, Japan

Koki Mimura

Advanced Neuroimaging Center, National Institutes for Quantum Science and Technology,
Chiba, Japan
ORCID iD: [0000-0001-7069-6558](https://orcid.org/0000-0001-7069-6558)

Kei Oyama

Advanced Neuroimaging Center, National Institutes for Quantum Science and Technology,
Chiba, Japan

Naohisa Miyakawa

Advanced Neuroimaging Center, National Institutes for Quantum Science and Technology,
Chiba, Japan

Yuki Hori

Advanced Neuroimaging Center, National Institutes for Quantum Science and Technology,
Chiba, Japan

Haruhiko Iwaoki

Advanced Neuroimaging Center, National Institutes for Quantum Science and Technology,
Chiba, Japan

Katsushi Kumata

Department of Advanced Nuclear Medicine Sciences, National Institutes for Quantum Science
and Technology, Chiba, Japan

Ming-Rong Zhang

Department of Advanced Nuclear Medicine Sciences, National Institutes for Quantum Science
and Technology, Chiba, Japan

Masahiko Takada

Advanced Neuroimaging Center, National Institutes for Quantum Science and Technology,
Chiba, Japan

Makoto Higuchi

Advanced Neuroimaging Center, National Institutes for Quantum Science and Technology,
Chiba, Japan

Takafumi Minamimoto

Advanced Neuroimaging Center, National Institutes for Quantum Science and Technology,
Chiba, Japan

ORCID iD: [0000-0003-4305-0174](https://orcid.org/0000-0003-4305-0174)

For correspondence: minamimoto.takafumi@qst.go.jp

Editors

Reviewing Editor

Kristine Krug

Otto-von-Guericke University Magdeburg, Magdeburg, Germany

Senior Editor

Joshua Gold

University of Pennsylvania, Philadelphia, United States of America

Reviewer #1 (Public review):

Summary:

Inhibitory hM4Di and excitatory hM3Dq DREADDs are currently the most commonly utilized chemogenetic tools in the field of nonhuman primate research, but there is a lack of available information regarding the temporal aspects of virally-mediated DREADD expression and function. Nagai et al. investigated the longitudinal expression and efficacy of DREADDs to modulate neuronal activity in the macaque model. The authors demonstrate that both hM4Di and hM3Dq DREADDs reach peak expression levels after approximately 60 days and that stable expression was maintained for up to two years for hM4Di and at least one year for hM3Dq DREADDs. During this period, DREADDs effectively modulated neuronal activity, as evidenced by a variety of measures, including behavioural testing, functional imaging, and/or electrophysiological recording. Notably, some of the data suggest that DREADD expression may decline after two-three years. This is a novel finding and has important implications for the utilization of this technology for long-term studies, as well as its potential therapeutic applications. Lastly, the authors highlight that peak DREADD expression may be significantly influenced by the presence of fused or co-expressed protein tags, emphasizing the importance of careful design and selection of viral constructs for neuroscientific research. This study represents a critical step in the field of chemogenetics, setting the scene for future development and optimization of this technology.

Strengths:

The longitudinal approach of this study provides important preliminary insights into the long-term utility of chemogenetics, which has not yet been thoroughly explored.

The data presented are novel and inclusive, relying on well-established in vivo imaging methods as well as behavioral and immunohistochemical techniques. The conclusions made by the authors are generally supported by a combination of these techniques. In particular, the utilization of in vivo imaging as a non-invasive method is translationally relevant and likely to make an impact in the field of chemogenetics, such that other researchers may adopt this method of longitudinal assessment in their own experiments. Rigorous standards have been applied to the datasets, and the appropriate controls have been included where possible.

The number of macaque subjects (20) from which data was available is also notable. Behavioral testing was performed in 11 subjects, FDG-PET in 5, electrophysiology in 1, and [11C]DCZ-PET in 15. This is an impressive accumulation of work that will surely be appreciated by the growing community of researchers using chemogenetics in nonhuman primates.

The implication that chemogenetic effects can be maintained for up to 1.5-2 years, followed by a gradual decline beyond this period, is an important development in knowledge. The limited duration of DREADD expression may present an obstacle in the translation of chemogenetic technology as a potential therapeutic tool, and it will be of interest for researchers to explore whether this limitation can be overcome. This study therefore represents a key starting point upon which future research can build.

Weaknesses:

None.

<https://doi.org/10.7554/eLife.105815.2.sa3>

Reviewer #2 (Public review):

Summary:

This paper reports histological, PET imaging, functional and behavioural data evaluating the longevity of AAV2 infection in multiple brain areas of macaques in the context of DREADD experiments. The central aim is to provide unprecedented information about how long the expression of HM4di or HM3dq receptors are expressed and efficient in modulating brain functions after vector injections. The data show peak expression after 40 to 60 days of vector injection, and stable expressions for up to 1.5 years for hM4di, and that hM3dq remained mostly at 75% of peak after a year, declining to 50% after 2 years. DREADDs effectively modulated neuronal activity and behaviour for approximately two years, evaluated with behavioural testings, neural recordings or FDG-PET. A statistical evaluation revealed that vector titers, DREADD type and tags contribute to the measured peak level of DREADD expression.

The article presents a thorough discussion of the limitations and specificities of chemogenetic approaches in monkeys.

Strength:

These are unique data, in non-human primate (NHP), an animal model that not only features physiological and immunological characteristics similar to humans, but also contributes to neurobiological functional studies over long timescales with experiments spanning months or years. This evaluation of long-term efficacy of DREADDs will be very important for all laboratories using chemogenetics in NHP but also for future use of such approach in experimental therapies. The longevity estimates are based on multiple approaches including behavioural and neurophysiological, thus providing information on functional efficacy of DREADD expression.

Performing such evaluation requires specific tools like PET imaging that very few monkey labs have access to. This study was done by the laboratory that has developed the radiotracer c11-DCZ, used here, a radiotracer binding selectively to DREADDs and providing, using PET, quantitative in vivo measures of DREADD expression. This study and its data should thus be a reference in the field, providing estimates to plan future chemogenetic experiments.

Publishing databases of experimental outcomes in NHP DREADD experiments is crucial for the community because such experiments are rare, expensive and long. It contributes to refining experiments and reducing the number of animals overall used in the domain.

Weaknesses:

This study is a meta-analysis of several experiments performed in one lab. The good side is that it combined a large amount of data that might not have been published individually; the down side is that all things were not planned and equated, creating a lot of unexplained variances in the data. However, this was judiciously used by the authors to provide very relevant information. One might think that organized multi-centric experiments planned using the knowledge acquired here, will provide help testing more parameters, including some related to inter-individual variability, and particular genetic constructs.

<https://doi.org/10.7554/eLife.105815.2.sa2>

Reviewer #3 (Public review):

Summary

This manuscript, from the developers of the novel DREADD-selective agonist DCZ (Nagai et al., 2020), utilizes a unique dataset where multiple PET scans in a large number of monkeys, including baseline scans before AAV injection, 30-120 days post-injection, and then periodically over the course of the prolonged experiments, were performed to assess short- and long-term dynamics of DREADD expression in vivo, and to associate DREADD expression with the efficacy of manipulating the neuronal activity or behavior. The goal was to provide critical insights into practicality and design of multi-year studies using chemogenetics, and to elucidate factors affecting expression stability.

Strengths are systematic quantitative assessment of the effects of both excitatory and inhibitory DREADDs, quantification of both the short-term and longer-term dynamics, a wide range of functional assessment approaches (behavior, electrophysiology, imaging), and assessment of factors affecting DREADD expression levels, such as serotype, promoter, titer (concentration), tag, and DREADD type.

These findings will undoubtedly have a very significant impact on the rapidly growing, but still highly challenging field of primate chemogenetic manipulations. As such, the work represents an invaluable resource for the community.

<https://doi.org/10.7554/eLife.105815.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Overall, the conclusions of the paper are mostly supported by the data but may be overstated in some cases, and some details are also missing or not easily recognizable within the figures. The provision of additional information and analyses would be valuable to the reader and may even benefit the authors' interpretation of the data.

We thank the reviewer for the thoughtful and constructive feedback. We are pleased that the reviewer found the overall conclusions of our paper to be well supported by the data, and we

appreciate the suggestions for improving figure clarity and interpretive accuracy. Below, we address each point with corresponding revisions.

The conclusion that DREADD expression gradually decreases after 1.5-2 years is only based on a select few of the subjects assessed; in Figure 2, it appears that only 3 hM4Di cases and 2 hM3Dq cases are assessed after the 2-year timepoint. The observed decline appears consistent within the hM4Di cases, but not for the hM3Dq cases (see Figure 2C: the AAV2.1-hSyn-hM3Dq-IRES-AcGFP line is increasing after 2 years.)

We agree that our interpretation should be stated more cautiously, given the limited number of cases assessed beyond the two-year timepoint. In the revised manuscript, we have clarified in the Results that the observed decline is based on a subset of animals. We have also included a text stating that while a consistent decline was observed in hM4Di-expressing monkeys, the trajectory for hM3Dq expression was more variable with at least one case showing an increased signal beyond two years.

Revised Results section:

Lines 140, “hM4Di expression levels remained stable at peak levels for approximately 1.5 years, followed by a gradual decline observed in one case after 2.5 years, and after approximately 3 years in the other two cases (Figure 2B, a and e/d, respectively). Compared with hM4Di expression, hM3Dq expression exhibited greater post-peak fluctuations. Nevertheless, it remained at ~70% of peak levels after about 1 year. This post-peak fluctuation was not significantly associated with the cumulative number of DREADD agonist injections (repeated-measures two-way ANOVA, main effect of activation times, $F_{(1,6)} = 5.745$, $P = 0.054$). Beyond 2 years post-injection, expression declined to ~50% in one case, whereas another case showed an apparent increase (Figure 2C, c and m, respectively).”

Given that individual differences may affect expression levels, it would be helpful to see additional labels on the graphs (or in the legends) indicating which subject and which region are being represented for each line and/or data point in Figure 1C, 2B, 2C, 5A, and 5B. Alternatively, for Figures 5A and B, an accompanying table listing this information would be sufficient.

We thank the reviewer for these helpful suggestions. In response, we have revised the relevant figures (Fig. 1C, 2B, 2C, and 5) as noted in the “Recommendations for the authors”, including simplifying visual encodings and improving labeling. We have also updated Table 2 to explicitly indicate the animal ID and brain regions associated with each data point shown in the figures.

While the authors comment on several factors that may influence peak expression levels, including serotype, promoter, titer, tag, and DREADD type, they do not comment on the volume of injection. The range in volume used per region in this study is between 2 and 54 microliters, with larger volumes typically (but not always) being used for cortical regions like the OFC and dlPFC, and smaller volumes for subcortical regions like the amygdala and putamen. This may weaken the claim that there is no significant relationship between peak expression level and brain region, as volume may be considered a confounding variable. Additionally, because of the possibility that larger volumes of viral vectors may be more likely to induce an immune response, which the authors suggest as a potential influence on transgene expression, not including volume as a factor of interest seems to be an oversight.

We thank the reviewer for raising this important issue. We agree that injection volume could act as a confounding variable, particularly since larger volumes were used in only handheld cortical injections. This overlap makes it difficult to disentangle the effect of volume from

those of brain region or injection method. Moreover, data points associated with these larger volumes also deviated when volume was included in the model.

To address this, we performed a separate analysis restricted to injections delivered via microinjector, where a comparable volume range was used across cases. In this subset, we included injection volume as additional factor in the model and found that volume did not significantly impact peak expression levels. Instead, the presence of co-expressed protein tags remained a significant predictor, while viral titer no longer showed a significant effect. These updated results have replaced the originals in the revised Results section and in the new Figure 5. We have also revised the Discussion to reflect these updated findings.

The authors conclude that vectors encoding co-expressed protein tags (such as HA) led to reduced peak expression levels, relative to vectors with an IRES-GFP sequence or with no such element at all. While interesting, this finding does not necessarily seem relevant for the efficacy of long-term expression and function, given that the authors show in Figures 1 and 2 that peak expression (as indicated by a change in binding potential relative to non-displaced radioligand, or ΔBP_{ND}) appears to taper off in all or most of the constructs assessed. The authors should take care to point out that the decline in peak expression should not be confused with the decline in longitudinal expression, as this is not clear in the discussion; i.e. the subheading, "Factors influencing DREADD expression," might be better written as, "Factors influencing peak DREADD expression," and subsequent wording in this section should specify that these particular data concern peak expression only.

We appreciate this important clarification. In response, we have revised the title to "Protein tags reduce peak DREADD expression levels" in the Results section and "Factors influencing peak DREADD expression levels" in the Discussion section. Additionally, we specified that our analysis focused on peak ΔBP_{ND} values around 60 days post-injection. We have also explicitly distinguished these findings from the later-stage changes in expression seen in the longitudinal PET data in both the Results and Discussion sections.

Reviewer #1 (Recommendations for the authors):

(1) Will any of these datasets be made available to other researchers upon request?

All data used to generate the figures have been made publicly available via our GitHub repository (<https://github.com/minamimoto-lab/2024-Nagai-LongitudinalPET.git>). This has been stated in the "Data availability" section in the revised manuscript.

(2) Suggested modifications to figures:

a) In Figures 2B and C, the inclusion of "serotype" as a separate legend with individual shapes seems superfluous, as the serotype is also listed as part of the colour-coded vector

We agree that the serotype legend was redundant since this information is already included in the color-coded vector labels. In response, we have removed the serotype shape indicators and now represent the data using only vector-construct-based color coding for clarity in Figure 2B and C.

b) In Figures 3A and B, it would be nice to see tics (representing agonist administration) for all subjects, not just the two that are exemplified in panels C-D and F-H. Perhaps grey tics for the non-exemplified subjects could be used.

In response, we have included black and white ticks to indicate all agonist administration across all subjects in Figure 3A and B, with the type of agonist clearly specified.

c) In Figure 4C, a Nissl- stained section is said to demonstrate the absence of neuronal loss at the vector injection sites. However, if the neuronal loss is subtle or widespread, this might not be easily visualized by Nissl. I would suggest including an additional image from the same section, in a non-injected cortical area, to show there is no significant difference between the injected and non-injected region.

To better demonstrate the absence of neuronal loss at the injection site, we have included an image from the contralateral, non-injected region of the same section for comparison (Fig. 4C).

d) In Figure 5A: is it possible that the hM3Dq construct with a titer of 5×10^{13} gc/ml is an outlier, relative to the other hM3Dq constructs used?

We thank the reviewer for raising this important observation. To evaluate whether the high-titer constructs represented a statistical outlier that might artifactually influence the observed trends, we performed a permutation-based outlier analysis. This assessment identified this point in question, as well as one additional case (titer 4.6×10^{13} gc/ml, #255, L_Put), as significant outlier relative to the distribution of the dataset.

Accordingly, we excluded these two data points from the analysis. Importantly, this exclusion did not meaningfully alter the overall trend or the statistical conclusions—specifically, the significant effect of co-expressed protein tags on peak expression levels remain robust. We have updated the Methods section to describe this outlier handling and added a corresponding note in the figure legend.

Reviewer #2 (Public review):

Weaknesses

This study is a meta-analysis of several experiments performed in one lab. The good side is that it combined a large amount of data that might not have been published individually; the downside is that all things were not planned and equated, creating a lot of unexplained variances in the data. This was yet judiciously used by the authors, but one might think that planned and organized multicentric experiments would provide more information and help test more parameters, including some related to inter-individual variability, and particular genetic constructs.

We thank the reviewer for bringing this important point to our attention. We fully acknowledge that the retrospective nature of our dataset—compiled from multiple studies conducted within a single laboratory—introduces variability related to differences in injection parameters and scanning timelines. While this reflects the practical realities and constraints of long-term NHP research, we agree that more standardized and prospectively designed studies would better control such source of variances. To address this, we have added the following statement to the "Technical consideration" section in Discussion:

Lines 297, "This study included a retrospective analysis of datasets pooled from multiple studies conducted within a single laboratory, which inherently introduced variability across injection parameters and scan intervals. While such an approach reflects real-world practices in long-term NHP research, future studies, including multicenter efforts using harmonized protocols, will be valuable for systematically assessing inter-individual differences and optimizing key experimental parameters."

Reviewer #2 (Recommendations for the authors):

I just have a few minor points that might help improve the paper:

(1) Figure 1C y-axis label: should add ΔBP_{ND} in parentheses for clarity.

We have added “ ΔBP_{ND} ” to the y-axis label for clarity.

The choice of a sigmoid curve is the simplest clear fit, but it doesn't really consider the presence of the peak described in the paper. Would there be a way to fit the dynamic including fitting the peak?

We agree that using a simple sigmoid curve for modeling expression dynamics is a limitation. In response to this and a similar comment from Reviewer #3, we tested a double logistic function (as suggested) to see if it better represented the rise and decline pattern. However, as described below, the original simple sigmoid curve was a better fit for the data. We have included a discussion regarding this limitation of this analysis. See Reviewer #3 recommendations (2) for details.

The colour scheme in Figure 1C should be changed to make things clearer, and maybe use another dimension (like dotted lines) to separate hM4Di from hM3Dq.

We have improved the visual clarity of Figure 1C by modifying the color scheme to represent vector construct and using distinct line types (dashed for hM4Di and solid for hM3Dq data) to separate DREADD type.

(2) Figure 2

I don't understand how the referencing to 100 was made: was it by selecting the overall peak value or the peak value observed between 40 and 80 days? If the former then I can't see how some values are higher than the peak. If the second then it means some peak values occurred after 80 days and data are not completely re-aligned.

We thank the reviewer for the opportunity to clarify this point. The normalization was based on the peak value observed between 40–80 days post-injection, as this window typically captured the peak expression phase in our dataset (see Figure 1). However, in some long-term cases where PET scans were limited during this period—e.g., with one scan performing at day 40—it is possible that the actual peak occurred later. Therefore, instances where ΔBP_{ND} values slightly exceeded the reference peak at later time points likely reflect this sampling limitation. We have clarified this methodological detail in the revised Results section to improve transparency.

The methods section mentions the use of CNO but this is not in the main paper which seems to state that only DCZ was used: the authors should clarify this

Although DCZ was the primary agonist used, CNO and C21 were also used in a few animals (e.g., monkeys #153, #221, and #207) for behavioral assessments. We have clarified this in the Results section and revised Figure 3 to indicate the specific agonist used for each subject. Additionally, we have updated the Methods section to clearly specify the use and dosage of DCZ, CNO, and C21, to avoid any confusion regarding the experimental design.

Reviewer #3 (Public review):

Minor weaknesses are related to a few instances of suboptimal phrasing, and some room for improvement in time course visualization and quantification. These would be easily

addressed in a revision.

These findings will undoubtedly have a very significant impact on the rapidly growing but still highly challenging field of primate chemogenetic manipulations. As such, the work represents an invaluable resource for the community.

We thank the reviewer for the positive assessment of our manuscript and for the constructive suggestions. We address each comment in the following point-by-point responses and have revised the manuscript accordingly.

Reviewer #3 (Recommendations for the authors):

(1) Please clarify the reasoning was, behind restricting the analysis in Figure 1 only to 7 monkeys with subcortical AAV injection?

We focused the analysis shown in Figure 1 on 7 monkeys with subcortical AAV injections who received comparative injection volumes. These data were primary part of vector test studies, allowing for repeated PET scans within 150 days post-injection. In contrast, monkeys with cortical injections—including larger volumes—were allocated to behavioral studies and therefore were not scanned as frequently during the early phase. We will clarify this rationale in the Results section.

(2) Figure 1: Not sure if a simple sigmoid is the best model for these, mostly peaking and then descending somewhat, curves. I suggest testing a more complex model, for instance, double logistic function of a type $f(t) = a + b/(1+\exp(-c(t-d))) - e/(1+\exp(-g*(t-h)))$, with the first logistic term modeling the rise to peak, and the second term for partial decline and stabilization*

We appreciate the reviewer's thoughtful suggestion to use a double logistic function to better model both the rising and declining phases of the expression curve. In response to this and similar comments from Reviewer #1, we tested the proposed model and found that, while it could capture the peak and subsequent decline, the resulting fit appeared less biologically plausible (See below). Moreover, model comparison using BIC favored the original simple sigmoid model (BIC = 61.1 vs. 62.9 for the simple and double logistic model, respectively). This information has been included in the revised figure legend for clarity.

Given these results, we retained the original simple sigmoid function in the revised manuscript, as it provides a sufficient and interpretable approximation of the early expression trajectory—particularly the peak expression-time estimation, which was the main purpose of this analysis. We have updated the Methods section to clarify our modeling and rationale as follows:

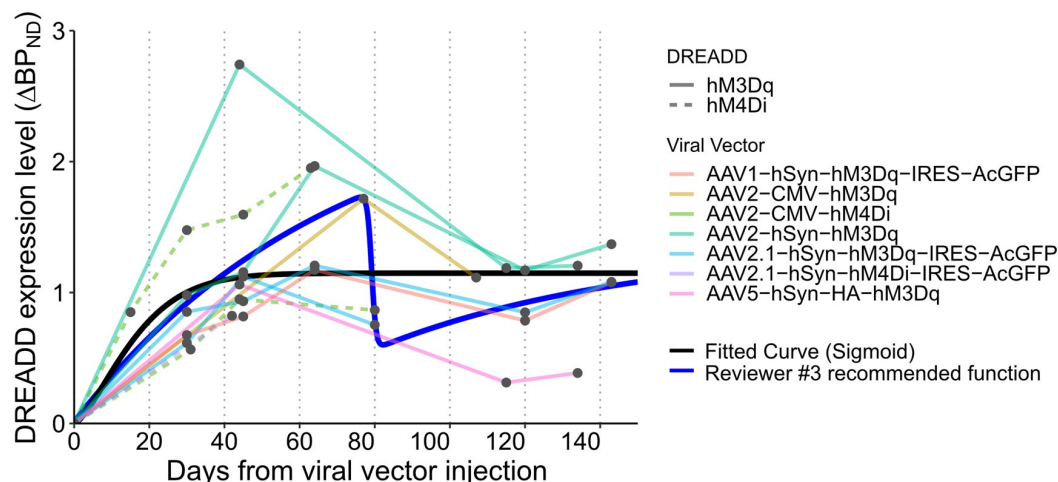
Lines 530, "To model the time course of DREADD expression, we used a single sigmoid function, referencing past in vivo fluorescent measurements (Diester et al., 2011). Curve fitting was performed using least squares minimization. For comparison, a double logistic function was also tested and evaluated using the Bayesian Information Criterion (BIC) to assess model fit."

We also acknowledge that a more detailed understanding of post-peak expression changes will require additional PET measurements, particularly between 60- and 120-days post-injection, across a larger number of animals. We have included this point in the revised Discussion to highlight the need for future work focused on finer-grained modeling of expression decline:

Lines 317, "Although we modeled the time course of DREADD expression using a single sigmoid function, PET data from several monkeys showed a modest decline following the peak. While the sigmoid model captured the early-phase dynamics and offered a reliable

estimate of peak timing, additional PET scans—particularly between 60- and 120-days post-injection—will be essential to fully characterize the biological basis of the post-peak expression trajectories.”

Author response image 1.



(3) Figure 2: It seems that the individual curves are for different monkeys, I counted 7 in B and 8 in C, why "across 11 monkeys"? Were there several monkeys both with hM4Di and hM3Dq? Does not look like that from Table 1. Generally, I would suggest associating specific animals from Tables 1 and 2 to the panels in Figures 1 and 2.

Some animals received multiple vector types, leading to more curves than individual subjects. We have revised the figure legends and updated Table 2 to explicitly relate each curve with the specific animal and brain region.

(4) I also propose plotting the average of (interpolated) curves across animals, to convey the main message of the figure more effectively.

We agree that plotting the mean of the interpolated expression curves would help convey the group trend. We added averaged curves to Figure 2BC.

(5) Similarly, in line 155 "We assessed data from 17 monkeys to evaluate ... Monkeys expressing hM4Di were assessed through behavioral testing (N = 11) and alterations in neuronal activity using electrophysiology (N = 2)..." - please explain how 17 is derived from 11, 2, 5 and 1. It is possible to glean from Table 1 that it is the calculation is 11 (including 2 with ephys) + 5 + 1 = 17, but it might appear as a mistake if one does not go deep into Table 1.

We have clarified in both the text and Table 1 that some monkeys (e.g., #201 and #207) underwent both behavioral and electrophysiological assessments, resulting in the overlapping counts. Specifically, the dataset includes 11 monkeys for hM4Di-related behavior testing (two of which underwent electrophysiology testing), 5 monkeys assessed for hM3Dq with FDG-PET, and 1 monkey assessed for hM3Dq with electrophysiology, totaling 19 assessments across 17 monkeys. We have revised the Results section to make this distinction more explicit to avoid confusion, as follows:

Lines 164, "Monkeys expressing hM4Di (N = 11) were assessed through behavioral testing, two of which also underwent electrophysiological assessment. Monkeys expressing hM3Dq (N

= 6) were assessed for changes in glucose metabolism via [^{18}F]FDG-PET (N = 5) or alterations in neuronal activity using electrophysiology (N = 1)."

(6) Line 473: "These stock solutions were then diluted in saline to a final volume of 0.1 ml (2.5% DMSO in saline), achieving a dose of 0.1 ml/kg and 3 mg/kg for DCZ and CNO, respectively." Please clarify: the injection volume was always 0.1 ml? then it is not clear how the dose can be 0.1 ml/kg (for a several kg monkey), and why DCZ and CNO doses are described in ml/kg vs mg/kg?

We thank the reviewer for pointing out this ambiguity. We apologize for the oversight and also acknowledge that we omitted mention of C21, which was used in a small number of cases. To address this, we have revised the "Administration of DREADD agonist" section of the Methods to clearly describe the preparation, the volume, and dosage for each agonist (DCZ, CNO, and C21) as follows:

Lines 493, "Deschloroclozapine (DCZ; HY-42110, MedChemExpress) was the primary agonist used. DCZ was first dissolved in dimethyl sulfoxide (DMSO; FUJIFILM Wako Pure Chemical Corp.) and then diluted in saline to a final volume of 1 mL, with the final DMSO concentration adjusted to 2.5% or less. DCZ was administered intramuscularly at a dose of 0.1 mg/kg for hM4Di activation, and at 1–3 $\mu\text{g/kg}$ for hM3Dq activation. For behavioral testing, DCZ was injected approximately 15 min before the start of the experiment unless otherwise noted. Fresh DCZ solutions were prepared daily.

In a limited number of cases, clozapine-N-oxide (CNO; Toronto Research Chemicals) or Compound 21 (C21; Tocris) was used as an alternative DREADD agonist for some hM4Di experiments. Both compounds were dissolved in DMSO and then diluted in saline to a final volume of 2–3 mL, also maintaining DMSO concentrations below 2.5%. CNO and C21 were administered intravenously at doses of 3 mg/kg and 0.3 mg/kg, respectively."

(7) Figure 5A: What do regression lines represent? Do they show a simple linear regression (then please report statistics such as R-squared and p-values), or is it related to the linear model described in Table 3 (but then I am not sure how separate DREADDs can be plotted if they are one of the factors)?

We thank the reviewer for the insightful question. In the original version of Figure 5A, the regression lines represented simple linear fits used to illustrate the relationship between viral titer and peak expression levels, based on our initial analysis in which titer appeared to have a significant effect without any notable interaction with other factors (such as DREADD type).

However, after conducting a more detailed analysis that incorporated injection volume as an additional factor and excluded cortical injections and statistical outliers (as suggested by Reviewer #1), viral titer was no longer found to significantly predict peak expression levels. Consequently, we revised the figure to focus on the effect of reporter tag, which remained the most consistent and robust predictor in our model.

In the updated Figure 5, we have removed the relationship between viral titer and expression level with regression lines.

<https://doi.org/10.7554/eLife.105815.2.sa0>